Flight Test Report

Experimental activities at Pipistrel

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The present document is submitted in partial fulfilment of the evaluation requirements for the *Sperimentazione in Volo* (*Flight Testing*) course, M.Sc. Aeronautical Engineering, Politecnico di Milano.

Executive summary

With the following report, the candidates intend to illustrate the results obtained from the test campaign carried out during the flight testing experience in collaboration with the Slovenian aircraft manufacturing company Pipistrel. Following the design of the flight mission, which was agreed upon with the course instructor, the candidates played a crucial role in the field of flight experimentation, engaging with the very foundations of aeronautical certification and thereby enriching their knowledge base with essential insights into a highly specific and delicate domain.

The data acquired during the tests were subsequently cross-checked in the post-processing phase with the data provided by the company, thanks to the presence of sophisticated software installed on board the aircraft involved in the trials.

The initial section of the report is dedicated to the introduction of all the instruments employed for the execution of the experience, followed by a section describing the mission tests, which includes the results obtained and their comparison with the data available from the Pilot's Operating Handbook (POH).

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Acronyms

Acronym	Definition
AGL	Above Ground Level
AHRS	Artificial Horizon System
ALT	Altimeter
AOA	Angle of Attack
ASI	Airspeed Indicator
ASL	Above Sea Level
ATZ	Air Traffic Aerodrome
CAS	Calibrated Airspeed
CTR	Control Zone
EGT	Exhaust Gas Temperature
FTP	Flight Test Pilot
FTE	Flight Test Engineer
FTI	Flight Testing Instrumentation
GCS	Ground Control System
HSI	Horizontal Situation Indicator
IAS	Indicated Airspeed
MAP	Manifold Absolute Pressure
OAT	Outside Air Temperature
PFD	Primary Flight Display
RPM	Revolutions Per Minute
SOC	State of Charge
SOH	State of Health
STOL	Short Take-off and Landing
TAS	True Airspeed
VHF	Very High Frequency
VOR	Visual Flight Rules
WAAS	Wide Area Augmentation System

Symbols

Symbol	Definition
$\boldsymbol{V_{NE}}$	Never Exceed Speed
$\boldsymbol{V_{FE}}$	Maximum Flap Extended Speed
$\boldsymbol{V_{AE}}$	Max Full Airbrakes Speed
$\boldsymbol{V_{S}}$	Stall speed - clean (flaps 0)
$\boldsymbol{V_{S1}}$	Stall speed - take-off (flaps +1)
$\boldsymbol{V_{S0}}$	Stall speed - landing (flaps +2)
$\boldsymbol{V}_{\mathrm{trim}}$	Trim speed
$\boldsymbol{V_{R}}$	Rotation Speed
$\boldsymbol{V_{LOF}}$	Lift-Off Speed
$\boldsymbol{V_{5}}$	Airspeed at 50 ft AGL
$\boldsymbol{V_{TD}}$	Touchdown Speed
$\boldsymbol{S_{g}}$	Ground Roll Distance
$\boldsymbol{S_{TO}}$	Total Take-Off Distance
$\boldsymbol{S_{3}}$	Distance: obstacle - touchdown
$\boldsymbol{S_{4}}$	Distance: touchdown - stop
$\boldsymbol{S_{LND}}$	Total Landing Distance
$\boldsymbol{MTOW}$ / $\boldsymbol{MLW}$	Max Takeoff / Landing Weight
$\boldsymbol{M_{F}}$	Total usable fuel
$\boldsymbol{R}$	Range (LND @ 30% SOC)
$\boldsymbol{E}$	Endurance (LND @ 30% SOC)
$\boldsymbol{\zeta}$	Oscillation Damping Ratio
$\boldsymbol{T}$	Oscillation Period

Abbreviation	Definition
TO	Take-off
LND	Landing
GLD	Glider

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1 Introduction

1.1 General objectives

The objective of the flight test campaign was to verify and validate the performance, handling qualities, and operational characteristics of the Pipistrel SW121 Explorer and SW128 Electro aircrafts, in line with the agreed planning document. The missions were executed with similar flight profiles on both aircraft to ensure consistency and comparability.

1.2 Test aircraft

Both employed aircraft are classified as ultralight and are manufactured by Pipistrel, but they differ significantly in propulsion system and performance characteristics, aspects relevant to the specific tests to be conducted.

Explorer – SW121
The Pipistrel SW121 Explorer is a two-seat, high-wing, T-tail ultralight aircraft with fixed tricycle landing gear, powered by a 100 hp Rotax 912 ULS engine and equipped with a variable-pitch propeller. It is certified in the CS-LSA category with a maximum takeoff weight of 600 kg. The aircraft measures 6.42 m in length, with a wingspan of 10.70 m and a wing area of 9.51 m². The horizontal tail spans 2.18 m with an area of 1.39 m², while the vertical tail has an area of 1.24 m² and a height of 1.11 m. Its typical empty weight is 371 kg, providing a useful load of 229 kg, including up to 25 kg of baggage.

Velis Electro – SW128
The Pipistrel SW128 Velis Electro is a two-seat, high-wing, T-tail aircraft with fixed tricycle landing gear and a liquid-cooled electric motor rated at 57.6 kW, driving a fixed-pitch propeller. It is certified under EASA CS-LSA for VFR day operations, with a maximum take-off weight of 600 kg. Its dimensions are almost equal to the Explorer, with a length of 6.47 m, wingspan of 10.71 m, and wing area of 9.51 m². The horizontal tail spans 2.18 m with an area of 1.39 m², and the vertical tail has an area of 1.24 m² and a height of 1.11 m. The aircraft has a design empty weight of 428 kg and a useful load of 172 kg, with no baggage allowance.

1.3 Test instrumentation

The following tables gather all the testing instrumentation present onboard for both the SW121 and SW128. The instrumentation is divided between **native** and **additional**

Explorer - SW121

Sensor	Measured Data
Garmin GEA 24 EIS	Coolant temperature, Exhaust Gas Temperature (EGT), Manifold Absolute Pressure (MAP), oil pressure, oil temperature, fuel pressure, fuel flow, voltage, current (amperage), engine RPM. Sampling frequency: **10Hz**.
Garmin GSU 25C ADAHRS	Roll, pitch, yaw, magnetic heading, tri-axial acceleration, Outside Air Temperature (OAT), True Airspeed (TAS) and wind vector calculations. HSI (horizontal situation indicator). Utilises data from a remote magnetometer for enhanced accuracy. Sampling frequency: **10Hz**.
Garmin GAP 26	Total and static pressure, indicated airspeed, angle of attack. Sampling frequency: **10Hz**.

Mechanical Altimeter	Three-pointer barometric altimeter, independent of electrical power.
Garmin GMU 22	Measures the Earth's magnetic field to determine magnetic heading; operates in conjunction with the ADAHRS. Sampling frequency: **10Hz**.
Magnetic Compass	Indicates magnetic heading. Serves as an independent backup navigation aid.
Slip Indicator	Indicates turn coordination using a free-moving ball in a curved, oil-filled tube (passive instrument).
Turn Coordinator	Displays rate of turn and coordination; function integrated into the Primary Flight Display (PFD).
Garmin GNC 355A	VHF communications and GPS navigation with Wide Area Augmentation System (WAAS); supports RNAV procedures. Sampling frequency: **1Hz**.
Garmin GNC 255A	Provides VHF communication and receives navigation signals from VOR, Localiser (LOC), and Glideslope. Supports 8.33 kHz channel spacing.
Garmin GTX 345	Transmits aircraft position, altitude and velocity via ADS-B Out; receives traffic (TIS-B) and weather data (FIS-B) via ADS-B In.

TABLE 1: EXPLORER - SW121 NATIVE INSTRUMENTATION.

Sensor	Measured Data
FTI Controller + Display	Displays real-time flight test data acquired from onboard sensors. Event marking and Windowing (DATA BLOCK or TOP).
Potentiometer	Records angular displacement of elevator control surface.
GPS + RTK	Provides high-precision ground speed, altitude, and velocity vector components; also enables real-time telemetry transmission to the Ground Control Station (GCS).
Cockpit Camera + Voice	Recording of the mission of both audio and video from the plane cockpit.

TABLE 2: EXPLORER - SW121 ADDITIONAL INSTRUMENTATION.

Electro - SW128

Sensor	Function	Measured Data
EPSI570C	Integrated Avionics	Motor operating status, RPM, power and temperature. Power controller operating status, temperature. Battery system operating status, temperature, SOC and SOH. Auxiliary (14V) battery status and voltage. Sampling frequency: **16Hz**.
Kanardia Horis (AH)	Artificial Horizon	Provides aircraft attitude: pitch and roll, using internal sensors and LCD display. OAT measurement. Sampling frequency: **16Hz**.
Garmin Aera 660	GPS Navigator	Provides GNSS-based position, navigation, and moving map display. Sampling frequency: **1Hz**.
Kanardia VSI	Vertical Speed Indicator	Measures rate of climb or descent; displayed with needle and digital readout. Sampling frequency: **16Hz**.
Mechanical Altimeter	Barometric Altimeter	Indicates altitude based on local barometric pressure; conventional three-pointer.
Mechanical ASI	Airspeed Indicator	Measures indicated airspeed via Pitot-static pressure difference (knots).

Funke ATR833 COM	Radio Communication	Provides VHF voice communication; supports 8.33 / 25 kHz channel spacing.
Funke TRT800H	Mode S Transponder	Transmits aircraft identification, position, and altitude; includes altitude encoder.
Mechanical Compass	Magnetic Heading	Displays aircraft heading relative to magnetic north; liquid-filled, manual.
Slipball Winter	Turn and Slip Indicator	Indicates turn rate and coordination using an inclinometer and gyroscopic input.

TABLE 3: ELECTRO - SW128 NATIVE INSTRUMENTATION.

Sensor	Measured Data
FTI Controller + Display	Displays real-time flight test data acquired from onboard sensors. Event marking and Windowing (DATA BLOCK or TOP).
Potentiometer	Records angular displacement of elevator control surface.
GPS + RTK	Provides high-precision ground speed, altitude, and velocity vector components; also enables real-time telemetry transmission to the Ground Control Station (GCS).
Cockpit Camera + Voice	Recording of the mission of both audio and video from the plane cockpit.

TABLE 4: ELECTRO - SW128 ADDITIONAL INSTRUMENTATION.

1.5 Limitations

When planning a test mission, it is essential to account for all relevant constraints, including weather, airspace, borders, terrain, and aircraft limitations. The main factors considered in mission design and planning are outlined below.

Airspace

The airspace around Gorizia Airport is subject to several restrictions. The airport lies within an ATZ, requiring radio communication for entry and exit in accordance with LIPG procedures. Over the city of Gorizia, prohibited area P76 extends from the surface to 1,500 ft AGL.
Other relevant controlled zones include:

- **Zone 21 "Palmanova"**: non-regulated from SFC to 3,000 ft AMSL; controlled from 3,000 ft AMSL to FL85.
- **Ronchi CTR**: SFC to 1,500 ft AMSL, entry/exit coordinated with Ronchi TMA.
- **Udine ATZ**: access requires radio contact with Udine Tower on 119.055 MHz.

Crossing into Slovenia must be pre-planned and coordinated with Slovenian authorities.

Take-off and Landing

- Take-off permitted from runways 27 and 09 (aircraft < 1,500 kg MTOW).
- Landing allowed only on runway 27; runway 09 is prohibited.
- All ATZ entries/exits must be communicated.

Aircraft (POH)

The planned mission remained within aircraft operational limits, with flight safety as the priority. The main constraints useful for the performed missions, for both the SW121 Explorer and SW128 Velis Electro, are summarised in the following Table 5.

Symbol	Definition	Value SW121	Value SW128
VNE	Never Exceed Speed (3'000 ft DA)	156 KIAS	108 KIAS
$VS0$	Stall Speed (+2)	47 KIAS	46 KIAS
$Vs1$	Stall Speed (+1)	51 KIAS	46 KIAS
Vs	Stall Speed (0)	53 KIAS	53 KIAS
Vcw	Max Crosswind Speed	18 kn	18 kn
$MTOW$ / MLW	Max Takeoff / Landing Weight	600 kg	600 kg
CG @ min TOW	CG limits, min weight (427 kg)	25% MAC / 35% MAC	—
CG @ max TOW	CG limits, max weight (600 kg)	27.5% MAC / 35% MAC	—
fwd / aft CG	CG limitations with crew	—	25.2% MAC / 32.6% MAC

TABLE 5: SW121 & SW128 POH CONSTRAINTS.

1.6 Schedule

The flight test mission followed the sequence outlined below. Both aircrafts performed the same flight test sequence, apart from the difference in performing the wind determination (clovereaf).

1. Familiarization with FTI.
2. Briefed PIC and company FTE, confirming the planning.
3. Completion of all pre-flight checklists and data acquisition start.
4. Taxi to runway 27 and prepare for the TO.
5. Take-off performance testing technique from runway 27 at Gorizia Airport.
6. Exit the airport traffic pattern, climb en route, and leave the ATZ of Gorizia. Continue climbing to a pressure altitude of 3,000 ft.
5. Perform four phugoid test points (2 stick-free and 2 stick-fixed), trying to fly upwind and downwind switching from stick free to stick fixed points.
6. Perform a four-leg cloverleaf to determine wind conditions and prepare for stall tests. (**ONLY SW121 crew**).
7. Stall Testing. Six test points, 3 upwind, 3 downwind, aiming for build-up entry rates of 0.5 kn/s, 1 kn/s, 1.5 kn/s.
8. Turn back toward Gorizia Airport, enter the traffic pattern and prepare for the landing test sequence.
9. Landing performance testing according to the prescribed procedure.
10. Full-stop on runway and record the final set of test data.
11. Post-flight checklists performed.
12. Aircraft and FTI shut down.

2 Test and evaluation

The flight test initial and final conditions were noted and saved in order to be able to know exactly the configuration and the state of the two aircraft during the testing. Here in the tables below are summarised the recorded conditions of the aircraft in the Flight Test Card.

SW121	
Date	29/05/2025
PIC	Martin Vadnu
FTE	Filippo Coacci
Engine-ON – UTC	11:01
HOBBS-ON	244.5 h
Engine-OFF – UTC	11:47
HOBBS-OFF	245.2 h
Take-off – UTC	11:08
Landing – UTC	11:44

SW128	
Date	29/05/2025
PIC	Joze Kovacic
FTE	Josè Bertè
HOBBS-ON	140:41
HOBBS-OFF	141:28
Take-off – UTC	11:20
Landing – UTC	11:58
SOH- Motor-ON	100
SOH- Motor-OFF	100
SOC – Motor-ON	100
SOC – Motor-OFF	30

TABLE 6: TAKE-OFF FLIGHT TEST INFORMATION.

Loading Conditions

Following the loading conditions for both the aircraft. The FTI Assembly consisted of a display, a computer, cables and battery, and a GPS antenna. The rest of the loading masses for each test flight is reported in the tables below.

PIC	85 kg
FTE	75 kg
FTI Assembly	7.1 kg
Fuel Engine-ON	50 L
Fuel Engine-OFF	41.7 L

PIC	70 kg
FTE	85 kg
FTI Assembly	7.1 kg

TABLE 7: LANDING FLIGHT TEST INFORMATION.

Testing Overview

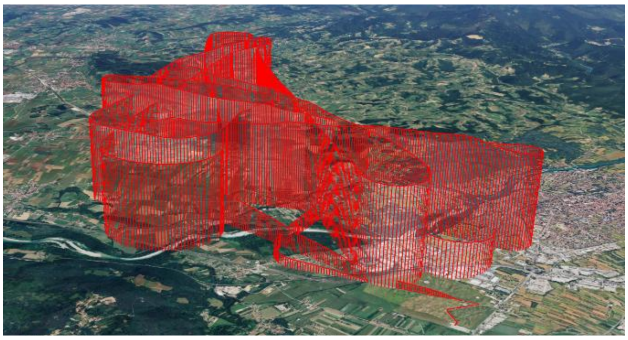


FIGURE 1: EXPLORER SW121 MISSION TRACK.

The test campaign was overall successful, and all data required for the analysis were correctly acquired. The flight tests were conducted between Gorizia LIPG ATZ and Zone 21 "Palmanova", in full compliance with local air traffic regulations. Following completion of the flights, the data from both aircraft were subsequently processed through the combined use of *Dewesoft* software, for the export of data and event markers, and *MATLAB* for post-processing and analysis. The following figures present the mission track of the Explorer SW128 and the plots of groundspeed and indicated airspeed throughout the mission, with event markers highlighting the various test points. Furthermore, it should be noted that the performance analyses presented in this report required reduction to a standard reference mass. While the SW128 maintained constant loading and mass throughout the test flight, being an electric aircraft, the SW121 required the determination of its instantaneous mass. For this purpose, the fuel consumption data provided by the on-board computer were integrated over time, enabling the calculation of both mass and centre of gravity position throughout the mission. The corresponding trend is shown in the graph below. The use of *Pipistrel's W&B Charts* for both planes was essential to obtain this type of data.

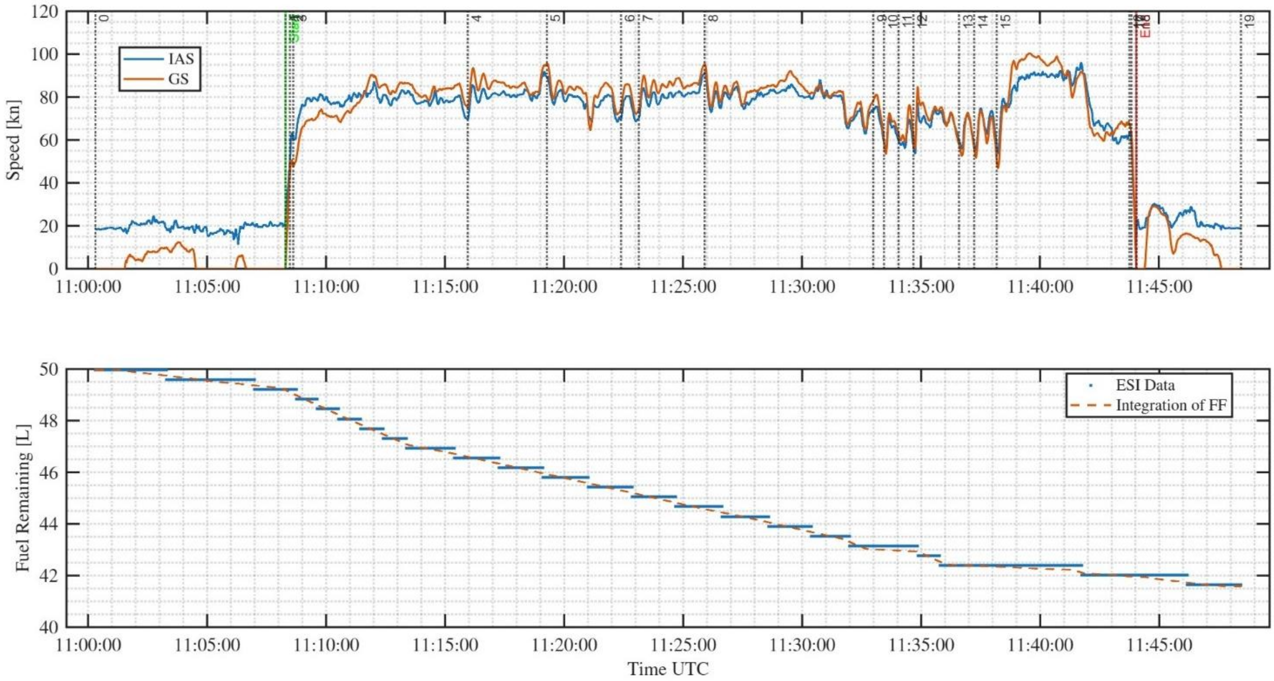


FIGURE 2: SW121MISSION AIRSPEED AND FUEL CONSUMPTION.

2.1 Take-off Performance

The objective of the test is to determine the minimum take-off distance (to overcome 50 ft obstacle) of the aircraft and to verify its compliance with EASA regulations, as well as with the performance data declared in the POH. The procedure was conducted in accordance with airfield regulations and in conformity with the methodologies prescribed in CS-23, paragraphs 23.53 and 23.59 (Amendment 4) [6], together with paragraph 4.4.2 of ASTM F2245–23 [4]. Main compliance with the certification includes:

a. The rotation speed $(V_R \geq 1.1 \cdot V_{S1})$.
b. Take-off configuration (flap +1 and most forward CG position).
c. The climb speed at 50 ft should be $(V_{50} \geq 1.3 \cdot V_{S1})$.

These regulatory references specify both the procedures for the determination of take-off performance and the correction methods required to account for environmental and operational factors, including wind conditions and runway slope.

2.1.1 Test method and conditions

Test Method

The take-off was conducted in the aircraft's standard take-off configuration (+1), as defined in the relevant sections of the Pilot's Operating Handbooks (POHs) [10][11]. Prior to commencing the sequence, all pertinent runway, atmospheric and aircraft parameters were recorded, since these are essential both for data correction and for test repeatability. The methodology applied is outlined below:

1. **Runway alignment:** The aircraft was aligned with the runway centreline.
2. **Condition logging:** Environmental and configuration parameters were recorded.
3. **Application of maximum power:** Maximum power was applied against full brakes.

4. **Brake release and timing start:** Once engine was stabilised, the brakes were released. The FTE simultaneously marked the event using both the event marker and the chronometer.
5. **Engine parameter recording:** MAP and RPM were recorded during the initial acceleration phase.
6. **Rotation speed call-out:** PIC called out the Rotation speed. FTP marked the event using both the chronometer and the event marker.
7. **Acceleration to climb speed:** The aircraft was accelerated to the climb speed prescribed by the certification basis.
8. **50-ft altitude call-out:** When the aircraft reached 50 ft AGL (referenced to the altimeter), the PIC called out the airspeed. The FTE simultaneously recorded the data.
9. **Final engine data:** Engine parameters were again recorded.

Instrumentation

Given the highly dynamic character of the take-off manoeuvre, the number of measured parameters was deliberately limited, in order to reduce crew workload and potential sources of error. Redundancy between manual and digital logging was maintained to ensure reliability and to allow cross-validation in case of anomalies or instrument failure. Table 8 reports the accuracy affecting the acquired data.

Measurement	Flight Card	Data - SW121	Accuracy	Data - SW128	Accuracy
Time	Chronometer	FTI GPS time	± 1 s	FTI GPS time	± 0.01 s
Speed	Displayed GS	FTI GPS speed	± 0.1 m/s	FTI GPS speed	± 0.01 m/s
Engine RPM	Displayed RPM	GEA 24 EIS	± 1 RPM	EPSI570C	± 1 RPM
Engine MAP / PW	Displayed MAP	GEA 24 EIS	± 3.4 Pa	EPSI570C	± 10 W
Height	Altimeter	GSU 25C + GPS FTI	± 1 m	AHRS + GPS FTI	± 1 m
Distance	None	GPS coordinates	± 1 m	FTI GPS coordinates	± 1 m

TABLE 8: TO MEASUREMENTS.

Test Conditions

Test conditions were maintained largely as planned, with the notable exception of wind. Although, in principle, wind influence can be corrected through reduction methods, the correction assumes a constant, non-turbulent component aligned with the runway. This assumption was not met, since the prevailing wind was turbulent and is therefore likely to have affected take-off performance.

	QNH [hPa]	OAT [°C]	Flaps	TOM [kg]	CG [% MAC]	Headwind [kn]	Runway slope [°]
SW121	1016	25	+1	583.8	28.66	12	negligible
SW128	1017	24	+1	583.4	29.20	9	negligible

TABLE 9: TO CONDITIONS.

Two remarks are necessary regarding the conditions:

1. Certification requirements specified that the take-off test be performed with the most forward CG. This requirement was not satisfied, as the actual loading corresponded to 27 % MAC (SW121) and 27.5 % MAC (SW128). This requirement was not met because of practical loading constraints.
2. The test was conducted in turbulent wind conditions. Although average values were recorded (Table 9), instantaneous wind variations during the manoeuvre remain unknown. These deviations are expected to have influenced the results and will be considered in the interpretation.

Reduction Methods

The following reduction methods were applied in the take-off performance assessment:

a. Mass correction to the POH reference value of 600 kg.
b. Wind correction factors, as specified in the POH and in literature [12][13].
c. Corrections for atmospheric conditions (pressure altitude and OAT).

The complete set of formulae used for data reduction and for correction of POH values is provided in Appendix B.

2.1.2 Test results, analysis and discussion

The take-off test was analysed using two complementary approaches. The first relied on the data collected in the flight test cards, while the second used the reduced dataset from the on-board FTI system. Comparing the two provided redundancy and cross-validation. The card-based analysis employed a simplified kinematic model of the take-off run, divided into ground roll and climb, each approximated as uniformly accelerated motion (details in Appendix B). The FTI-based approach, considered the reference, offered higher accuracy and avoided potential human errors in manual data acquisition. The dataset included aircraft attitude, air data (IAS, pressure altitude, OAT), and GPS-derived quantities (latitude, longitude, groundspeed, altitude) recorded continuously during the manoeuvre.

The processing of the take-off data was conducted as follows:

1. The starting time of the manoeuvre was identified from the groundspeed (GS) vector, with the ground roll defined as beginning when GS exceeded zero.
2. During the ground roll, aircraft pitch attitude remained approximately constant, until the rotation manoeuvre commenced and pitch began to increase.
3. Lift-off was identified as the point at which GPS altitude first became greater than zero. This point, expressed as GPS coordinates and elapsed time since the start of the take-off, marked the transition between the ground roll and the initial climb phase.
4. Aircraft calibrated airspeed (CAS) was monitored, as it is required for comparison with regulatory criteria.
5. The manoeuvre was considered complete once the aircraft reached an altitude of 50 ft above the runway surface (from GPS altitude).
6. At this point, GPS coordinates, airspeed, and elapsed time were extracted, enabling the determination of additional parameters such as engine and atmospheric conditions.
7. Ground roll distance and total distance to 50 ft were derived from the recorded GPS latitude and longitude.

Remark: To improve accuracy and ensure consistency, each event identified through FTI analysis was cross-checked against the corresponding entries in the flight test cards and the event markers recorded during the test.

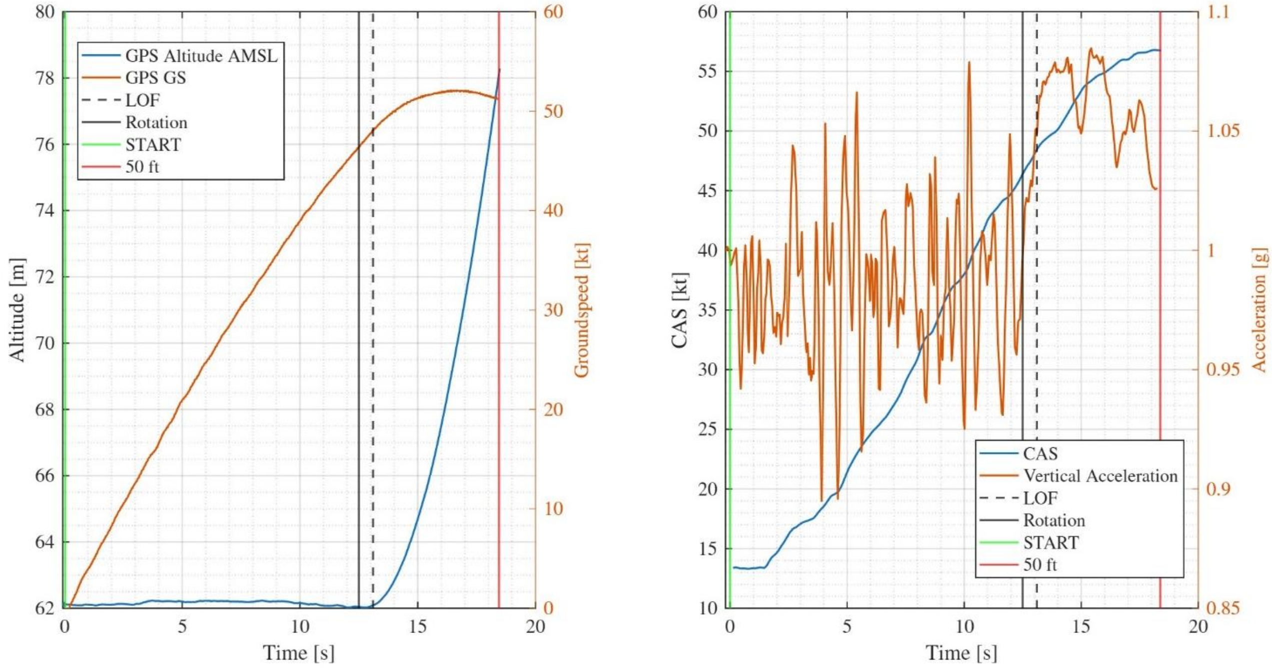


FIGURE 3: VELIS-SW128 TAKE-OFF MANOEUVRE.

From Figure 3, the full take-off manoeuvre can be visualised and analysed for the *Velis Electro-SW128*. The event markers for rotation and lift-off are observed to be consistent with the aircraft dynamics. The graphs clearly illustrate the identification of the key events and demonstrate how they can be analysed to determine the associated airspeed, distance and altitude values. The following table summarises the raw data extracted from the analysis of both aircraft.

	Airspeed CAS [kn]	SW121	SW128
FTI Data	**Rotation**	53	45.7
	LOF	54.7	48.1
	50 ft	62.8	56.8
FTC Data	**Rotation**	52	46
	LOF	54	48
	50 ft	63	58
POH Specs	**Rotation**	-	-
	LOF	47	50
	50 ft	59	61
Certification	**Rotation (1.1*Vs1)**	52.8	47.3
	50 ft (1.3*Vs1)	62.4	55.9

TABLE 10: TAKE-OFF TEST DATA.

The results show that the test was conducted in compliance with certification requirements and prescribed flight test techniques. Minor discrepancies between measured and expected values are plausibly due to slight delays in pilot control inputs during rotation and climb stabilisation, as well as instantaneous wind variations. These effects may have slightly increased both ground roll and total take-off distance. The close agreement between FTI and FTC data confirms the reliability of manual logging and demonstrates good execution by the FTEs.

Take-off Distance

The table below presents the corrected take-off distances. Both FTI and FTC data were reduced for the specific test conditions, notably wind, mass and power, using the methodology described in Appendix B. The POH values were interpolated to account for the runway elevation and OAT.

	FTI corrected data		FTC corrected data		POH data	
Distance [m]	**Ground roll**	**50 ft**	**Ground roll**	**50 ft**	**Ground roll**	**50 ft**
SW121	233	371	240	393	192	358
SW128	329	547	336	567	295	545

TABLE 11: TAKE-OFF CORRECTED RESULTS.

The results indicate that the measured performance underperformed relative to POH values by approximately 4% for SW121 and by less than 1% for SW128. Overall, the test results are therefore in close agreement with the manufacturer's stated performance.

The largest deviation is observed in the ground roll distances, where measured values consistently exceed those given in the POH. Several factors may explain this:

- **Pilot technique**: Even slight variations in rotation timing or acceleration management can affect ground roll significantly.
- **Wind variability**: The tests were conducted in turbulent wind, while POH corrections assume a steady, aligned component; this may have introduced additional scatter.
- **Surface conditions**: Differences between the actual runway condition and the assumptions underlying POH data (e.g. friction, slope tolerances) could also play a role.

Despite these discrepancies, the consistency between runs and the compliance with certification speeds suggest that the flight test technique was conservative and met the required standards. Consequently, the results could be considered valid for use in a certification context. Nevertheless, for formal certification purposes, multiple compliant runs would need to be conducted and statistically averaged to strengthen the reliability of the evidence.

Error Sources and data quality

The quality of the take-off data is influenced by both instrumentation and the dynamic nature of the manoeuvre. Manual entries on flight test cards are less precise than digital FTI outputs: a ±1 s timing error can result in ±15–20 m variation in ground roll. GPS-based distances are generally reliable (±1 m), though small errors can accumulate to ±5–10 m.

Other significant factors include:

- Turbulent wind conditions, which cannot be fully corrected with steady-wind methods.
- Pilot timing during rotation and pitch control.
- Aircraft loading and CG position, affecting rotation and ground roll.
- Runway surface unevenness and slope variations.

Overall, combined uncertainties may account for ±20–30 m in ground roll and ±30–40 m in total distance to 50 ft. Despite this, the agreement between FTI and FTC confirms that data quality is sufficient and within certification tolerances.

2.1.3 Conclusion and recommendations

The objective of determining the minimum take-off distance to 50 ft was met, with both aircraft achieving results closely aligned with the POH. After correction, the SW121 underperformed by about 4%, while the SW128 was within 1%, deviations consistent with normal variability and not indicative of performance deficiencies. Certification compliance was demonstrated: rotation and 50 ft speeds matched the 1.1 Vs1 and 1.3 Vs1 criteria, and the flap +1 configuration was correctly applied, though the most forward CG could not be achieved due to loading limitations, an effect expected to be minor.

The main discrepancies concerned ground roll distances, which exceeded POH values, likely due to wind turbulence, pilot technique at rotation, and runway surface effects. Pilots noted gusty winds made precise inputs difficult, adding variability. Instrumentation was adequate, with strong agreement between FTI and FTC data, though reliance on average wind values and lack of runway friction characterisation remain limitations.

Recommendations

- **R1-01**: Implement real-time wind measurement at the runway threshold (both direction and speed) to improve the accuracy of performance corrections.
- **R1-02**: Ensure test loading permits operation at the most forward CG position, to achieve full compliance with certification requirements.
- **R1-03**: Conduct multiple compliant runs under similar environmental conditions, in order to build a statistically significant dataset for certification use [13].
- **R1-04**: Provide additional pilot briefing and rehearsal focused on rotation timing, to minimise variability due to pilot technique in the dynamic phase of take-off.
- **R1-05**: Consider supplementary instrumentation (e.g. continuous wind sensors or runway surface monitoring) to reduce uncertainties associated with external conditions.

2.2 Longitudinal dynamic stability - Phugoid

The purpose of this test is to evaluate the aircraft’s long-period dynamic stability (phugoid mode) in accordance with CS-23 §23.181(c) [6] / CS-LSA[5] and ASTM F2245 §4.5.6 [4]. This mode is characterised by slow oscillations in pitch, altitude and airspeed, typically lightly damped, and acceptable provided it does not interfere with normal piloting tasks.

Dynamic stability is considered acceptable if:

- The oscillation period is < 15 s without visible divergence.
- Any instability develops with a doubling time > 55 s.
- The oscillation does not interfere with trimming, altitude keeping or glide slope tracking.
- A ±10% deviation in airspeed from the trimmed condition is used to initiate a representative phugoid oscillation.
- The aircraft should be disturbed from its trimmed flight path with a slow, deliberate elevator input, and then allowed to oscillate freely without further control input.

2.2.1 Test method and conditions

Each test point was preceded and concluded by a trim shot to record baseline flight conditions, including IAS, pressure altitude, OAT, RPM, MAP/Power, aircraft configuration, event markers, and, for SW121, weight. Oscillations were recorded using flight test cards and on-board instrumentation (FTI), with peaks and valleys logged alongside event markers and chronometer LAP times until the oscillation damped out.

Test sequence

1. Trim shot at 80 KIAS. FTE records baseline data.
2. Event marker + chronometer start.
3. Excitation: pitch down/up slowly (≥10 s) to reach ±10 KIAS from trim.
4. Stick-free: Control released to neutral after excitation. Lateral/directional corrections permitted.
5. Stick-fixed: Control returned and held at neutral throughout oscillation. Lateral/directional corrections permitted.
6. Oscillation allowed to evolve. FTE records peaks/valleys until damping.
7. Post-oscillation trim shot; FTE records final conditions.

Instrumentation

The set of parameters to be recorded during the test includes IAS, PA, OAT, pitch attitude. Redundancy between manual and digital logging was maintained to ensure reliability and to allow cross-validation in case of anomalies or instrument failure. Table 12 summarises the data gathered by the FTI e the relative accuracy.

Measurement	Flight Card	Data - SW128	Accuracy	Data - SW128	Accuracy
Time	FTI display	FTI GPS time	± 1 s	FTI GPS time	± 0.01 s
Airspeed	ASI	GSU 25C ADAHRS	± 0.1 kn	Kanardia AHRS	± 0.1 kn
PA	ALT	GSU 25C ADAHRS	± 1 ft	Kanardia AHRS	± 1 ft
OAT	EIS display	GSU 25C ADAHRS	± 1 °C	Kanardia AHRS	± 0.5 °C
Pitch Angle	Not recorded	GSU 25C ADAHRS	± 0.1 °	Kanardia AHRS	± 0.1 °
Fuel Remaining	EIS display	GEA 24 EIS	± 0.1 L	-	-

TABLE 12: PHUGOID INSTRUMENTATION.

Conditions

Test conditions were executed as planned, with the notable exception of wind. Flap position for both aircraft was set to 0. Stick-free phugoid tests were conducted first, followed by stick-fixed tests. The pilots-in-command adhered to the prescribed technique and performed the manoeuvres in accordance with the plan. However, turbulent wind significantly influenced the tests, causing some points to be missed, as the gusts disturbed aircraft attitude. Accurate estimation of wind conditions was not possible. The following table summarises the conditions for each test point.

	SW121					SW128		
	Hp [ft]	OAT [°C]	Mass [kg]	CG [% MAC]	Notes	Hp [ft]	OAT [°C]	Notes
Free-70 KIAS	4116	18	580.6	28.71	Light Turbulence	3005	17	Good
Free-90 KIAS	3960	16	579.9	28.72	Good	3015	18.5	Turbulence
Fixed-70 KIAS	4142	15	579.0	28.74	Light Turbulence	3002	17.5	Turbulence
Fixed-90 KIAS	4386	14	578.5	28.75	Good	3000	18	Good

TABLE 13: PHUGOID TEST CONDITIONS.

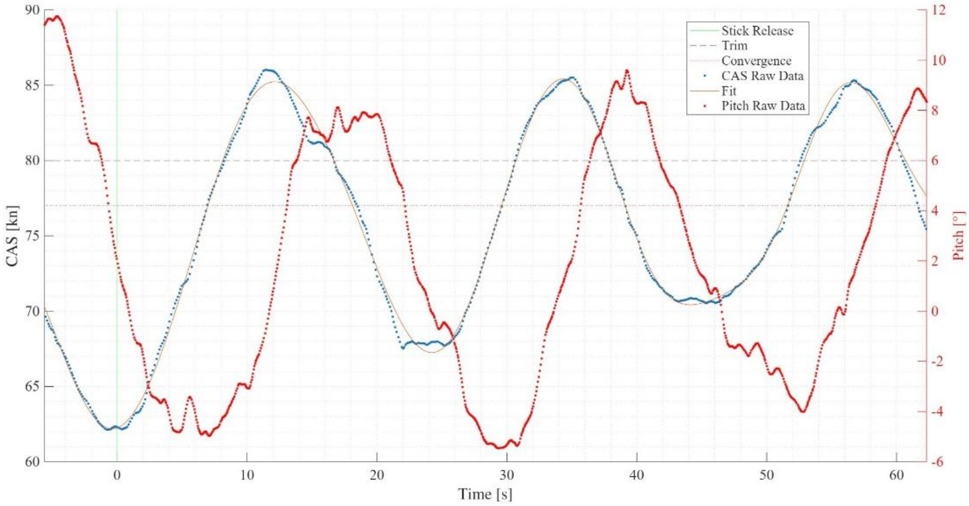
2.2.2 Test results, analysis and discussion

Both flight test cards and FTI data were used to analyse the phugoid motion. The results indicate that wind turbulence played a significant role in disturbing the oscillations. Nonetheless, valid test points were successfully obtained for both stick-fixed and stick-free test points of both aircraft. As summarised in Table 13, the first free and first fixed test points of the SW121, as well as the second free and first fixed points of the SW128, were strongly affected by turbulence according to both PIC and FTE observations. These points were therefore excluded from the analysis, as the deviations from nominal test conditions were too great for the results to be considered reliable.

The data reduction procedure for FTI was as follows:

1. Identification of the start time of each test point through the use of event markers.
2. Extraction of a time window of at least 15 seconds, in accordance with certification requirements [4][5][6], to allow reliable estimation of the oscillation period.
3. Plotting of relevant parameters, notably calibrated airspeed (CAS) and pitch attitude, against time.

4. Fitting of a sinusoidal function to the raw data in order to estimate the oscillation period and damping ratio.
5. Comparison of all valid test points against certification requirements.



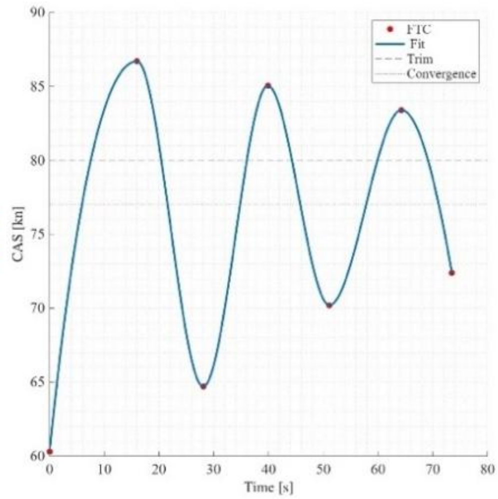


FIGURE 4: SW121 STICK-FIXED PHUGOID TEST POINT.

FIGURE 5: FTC SW121 STICK-FIXED TEST POINT.

An example of a valid test point for the SW128 is presented in Figure 4 and Figure 5, showing KCAS and pitch attitude versus time. The figure highlights the trim speed of 80 KCAS, the oscillation convergence, and the influence of wind on the motion. It also illustrates the phase shift between CAS and pitch attitude, revealing the inherent lag between body response and aerodynamic effects. The remaining plots are provided in Appendix D. The results are summarised in Table 14 for both aircraft and are compared with applicable certification requirements.

		T [s]	ζ [-]	Wind [% CAS trim]	Stability	Certification
SW121	**Free –70 KIAS**	27.0	0.00607	+ 1.6	Convergent	Acceptable
	Free – 90 KIAS	26.8	0.00532	+ 0.6	Convergent	Acceptable
	Fixed–70 KIAS	24.9	0.00537	- 1.4	Convergent	Acceptable
	Fixed – 90 KIAS	24.8	0.00615	- 1.1	Convergent	Acceptable
SW128	**Free –70 KIAS**	24.5	0.00701	- 2.2	Convergent	Acceptable
	Free – 90 KIAS	Failed	Failed	Failed	Failed	Not Acceptable
	Fixed–70 KIAS	Failed	Failed	Failed	Failed	Not Acceptable
	Fixed – 90 KIAS	25.4	0.00654	-2.1	Convergent	Acceptable

TABLE 14: PHUGOID TEST RESULTS.

All valid test points at the 80 KIAS trim speed demonstrated stable phugoid behaviour, sufficient to confirm that both aircraft, in the tested loading conditions, are longitudinally dynamically stable. The results derived from FTC data were consistent with the FTI-based analysis.

The oscillation period exceeded 15 seconds in all valid cases. While this does not strictly meet CS-23 criteria, ASTM F2245-23 [4] specifies that oscillations with a period greater than 15 seconds remain acceptable provided they converge towards equilibrium. Both aircraft exhibited such convergence and therefore comply with ASTM and CS-LSA certification requirements.

It should be noted that a complete certification demonstration would require repeating these tests across:

- The entire CG excursion range of each aircraft.

- The full speed envelope, from 1.1 *V*s up to maximum speed.
- Multiple repetitions to ensure statistical robustness.

Error Sources and data quality

The quality of the phugoid test data is affected by both instrumentation and the inherently long-period, low-damping nature of the manoeuvre. Because oscillations last 25–30 s, environmental disturbances and measurement uncertainties are amplified compared to shorter manoeuvres. Manual entries on flight test cards (e.g., marking peaks, IAS readings, timing) are less precise than FTI data: a ±1 s error in timing corresponds to ±3–4% uncertainty in period, while FTI timestamps ensure much higher accuracy.

Environmental conditions were a major source of error, with turbulence and thermals disrupting attitude and forcing the exclusion of certain points. Since phugoids require long, undisturbed flight paths, even modest turbulence can bias the damping ratio or mask natural decay. Aircraft-specific factors also played a role: CG variations influenced stability margins, with the SW121 subject to additional uncertainty from fuel consumption and post-flight mass estimation.

Overall, uncertainties from instrumentation, environment, and loading can account for ±2–4 s in period and ±10–20% in damping ratio. Nevertheless, consistency between FTI and FTC data shows that quality is adequate to demonstrate compliance, though further points across the full CG and speed range are recommended.

2.2.3 Conclusion and recommendations

The phugoid testing achieved its objective of evaluating the dynamic longitudinal stability of both the SW121 and SW128 at a representative cruise condition (80 KIAS, Flaps 0). Despite the influence of turbulence and the necessity to discard certain test points, the remaining data consistently showed convergent oscillations, in line with the ASTM F2245-23 allowance for long-period phugoids. Both aircraft can therefore be considered dynamically stable in the tested conditions.

From a qualitative standpoint, the pilots reported that turbulence had a strong perceptible impact on the motion, particularly masking the natural decay of oscillations. Quantitatively, wind turbulence was found to bias the damping ratio more strongly than the natural frequency: turbulence tends to increase or decrease the apparent damping, while the oscillation period remains largely unaffected. Additionally, uncertainty in CG position (particularly for the SW121, due to fuel consumption integration) may lead to either a slight underestimation or overestimation of stability margins, depending on the true shift of the static margin relative to the assumed value.

In terms of data acquisition and instrumentation, FTI proved essential for reliable oscillation analysis, while manual entries on the flight test card were subject to larger scatter. Overall, the objectives of the phugoid testing were met. Nevertheless, further testing across the entire CG range and at additional speeds within the flight envelope is recommended to ensure robustness of the conclusions.

Recommendations

- **R2-01** Conduct phugoid testing in calmer atmospheric conditions or at times of reduced thermal activity to minimise turbulence-induced damping bias.
- **R2-02** Improve wind measurement prior to and during testing (e.g., gust probes, high-frequency pitot-static data) to allow a more accurate decoupling of atmospheric effects from aircraft dynamics.
- **R2-03** Expand the test campaign to include phugoid manoeuvres across the full CG range and at multiple speeds (from 1.1 Vs up to Vne), to confirm stability throughout the operational envelope.
- **R2-04** Enhance mass and CG tracking on the SW121 by improving the accuracy of fuel flow integration, reducing bias in stability assessment.
- **R2-05** Provide tighter pilot technique standardisation (stick input and release timing) to reduce variability between repeated test points.

2.3 Stall Performance

The objective of the stall performance testing is to evaluate the stall behaviour of both the SW121 and SW128 in clean configuration (flaps 0) and to verify compliance with applicable certification requirements. In addition, the tests provide comparative insight into the aerodynamic characteristics of the two aircraft.

According to ASTM F2245-12d §4.4.1 [4] and CS-23 §23.201(b) [5][8], the stall speed Vs must be determined by flight testing with a rate of deceleration not exceeding 1 kn/s, throttle closed, maximum take-off weight, and the most aft centre of gravity. A stall is considered achieved when one of the following occurs: An uncontrollable nose-down pitching motion; A pitch-down triggered by an automatic protection system (e.g. stick pusher); A control deflection reaching its mechanical stop without further effect.

To demonstrate compliance, the following criteria are applied:

- Positive stick-force gradient up to stall (pull force required).
- Aircraft controllability in roll and yaw with unreversed inputs up to stall.
- Speed reduction rate during stall entry ≤ 1 kn/s.
- Extent of roll and/or yaw during stall and recovery documented.

Furthermore, from the CS-LSA [5] par. 4.5.7.3.1, 4.5.7.3.2, 4.5.7.3.3:

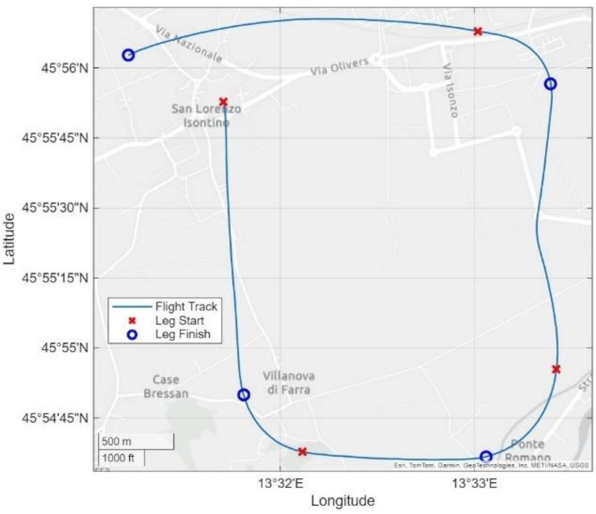
- Stall warning must be present.
- Stall warning must allow for sufficient time to recover aircraft before the stall.
- Stall warning may be furnished either through the inherent aerodynamic qualities (e.g. buffeting) of the aeroplane or by a device that clearly indicates the stall.

2.3.2 Wind Determination

Before the stall testing, the SW121 crew performed a wind cloverleaf manoeuvre to determine the wind vector. This was done for two main reasons:

1. Improving stall speed measurement accuracy by accounting for wind-induced IAS variations.
2. Ensuring consistent test conditions by orienting manoeuvres into the headwind or tailwind.

The procedure involved four straight legs at 90° heading increments, flown at the same altitude as the stall tests. On each leg, the pilot stabilised the aircraft while the FTE recorded track, GS, and TAS from the PFD. The results provided a quick wind estimate, allowing stall points to be executed both headwind and tailwind for improved interpretation. The table summarises the collected data and the calculated wind results.



GS [kn]	83	87	83	81	
Track [°]	175	92	360	263	
V wind [kn]	4.1	3.2	1.9	3.2	**3.1**
Wind dir [°]	268	286	268	246	**267**
					Average

FIGURE 6: SW121 - WIND DETERMINATION.

The results indicated an average wind of heading 269° and intensity 3.1 kn, leading the crew to perform stall tests on 270° (headwind) and 90° (tailwind) headings. The SW128 crew, constrained by limited endurance, skipped the cloverleaf and relied on pilot judgement for wind estimation. Although less precise, this proved acceptable, as the chosen headings matched the actual wind, underlining the value of PIC experience.

2.3.1 Test method and conditions

The stall tests were conducted in flap-retracted (0) configuration. The IAS-to-CAS correction applied in post-processing corresponds to this configuration.

Test sequence

The following steps were applied for each stall test point:

1. Trim shot at altitude: The pilot trimmed the aircraft at 1.5 × Vs [5], corresponding to about 76 KIAS. The FTE recorded the test conditions: PA, OAT, configuration, and fuel on board.
2. Entry to stall: Power was set to idle. The pilot executed decelerations at progressively increasing rates (0.5 kn/s, 1.0 kn/s, 1.5 kn/s) as part of a build-up approach. The FTE recorded the data.
3. Stall warning: The FTE marked the time of stall warning activation using the event marker.
4. Stall recognition: the pilot called aloud the IAS at stall onset, defined as described in the Stall Performance section of this report. The FTE simultaneously recorded IAS and UTC time.
5. Recovery: The pilot recovered promptly and repositioned the aircraft.

Each test point was performed both headwind and tailwind, with three entry rates in each direction. This allowed wind effects to be minimised and provided a basis for direct comparison.

Test Instrumentation

The set of parameters interesting for this test includes: IAS, PA, OAT, aircraft attitude angles (pitch, roll, yaw), aircraft attitude rates (pitch rate, roll rate, yaw rate), GPS-derived speeds, deceleration rate, engine power or MAP, engine RPM and longitudinal control force (based on pilot feedback).

Table 15 summarises the minimum required data, along with corresponding sensor codes, necessary to ensure the validity and consistency of the stall tests conducted on both the SW121 and SW128 aircraft.

Measurement	Flight Card	Data - SW121	Accuracy	Data - SW128	Accuracy
Time UTC	FTI display	FTI GPS time	± 1 s	FTI GPS time	± 0.01 s
Airspeed	ASI	GDU 470	± 0.1 kn	Kanardia AHRS	± 0.1 kn
PA	ALT	GSU 25C ADAHRS	± 1 ft	Kanardia AHRS	± 1 ft
OAT	PFD	GSU 25C ADAHRS	± 1 °C	Aera 660	± 0.5 °C
GS	FTI display	FTI GPS speed	± 0.1 kn	FTI GPS speed	± 0.1 kn
Entry Rate	FTI display	FTI GPS speed	± 0.5 kn/s	FTI GPS speed	± 0.5 kn/s
Fuel Remaining	EIS	GEA 24 EIS	± 0.1 L	-	-
RPM	EIS	GEA 24 EIS	± 1 RPM	EPSI570C	± 1 RPM
MAP / PW	EIS	GEA 24 EIS	± 3.4 Pa	EPSI570C	± 10 W
Pitch	-	GSU 25C ADAHRS	± 0.1 °	Kanardia AHRS	± 0.1 °
Roll	-	GSU 25C ADAHRS	± 0.1 °	Kanardia AHRS	± 0.1 °
Yaw	Compass	GSU 25C ADAHRS	± 0.1 °	Kanardia AHRS	± 0.1 °
Pitch rate	-	-	-	Kanardia AHRS	± 1e-4 rad/s
Roll rate	-	-	-	Kanardia AHRS	± 1e-4 rad/s
Yaw rate	-	-	-	Kanardia AHRS	± 1e-4 rad/s
Stick Force	Pilot feedback	-	-	-	-

TABLE 15: STALL TESTS INSTRUMENTATION.

Test Conditions

The test conditions were not ideal due to high wind turbulence. All the test points were performed accordingly to the planned technique and schedule. Some test point needed inevitably to be repeated due to excessive turbulence. Both aircrafts performed the test points in clean configuration (flap 0). Below the table summarizes the recorded conditions for each test point.

	HDG [°]	PA [ft]	OAT [°C]	Mass [kg]	CG [%MAC]	Notes
SW121	90	4884	14	576.8	28.77	Turbulence
	90	4085	14	576.8	28.77	Good
	90	3752	14	576.7	28.78	Good
	270	4778	13	576.2	28.78	Re-taken
	270	4102	13	576.2	28.78	Good
	270	3950	14	576.1	28.79	Good
SW128	77	2869	17.5	583.4	29.20	Good
	80	3150	17.5	583.4	29.20	Turbulence
	82	3150	17	583.4	29.20	Turbulence
	260	3165	17	583.4	29.20	Good
	260	3010	17.5	583.4	29.20	Good
	260	3063	17.5	583.4	29.20	Good

TABLE 16:CONDITIONS DURING STALL TESTS.

As we can see from Table 16, there were some test points that were strongly affected by turbulence. This resulted in inconsistent results, for example higher stalling speeds at higher deceleration rates, which is not in accordance with theory.

Deviations

It is important to note that the certification [4][5] require the stall test to be performed with at the maximum take-off weight and with the CG at the most unfavourable position. Both these requirements were not met due to loading constraints. In the results section will be possible to reduce the results based on the relative ass at each test point, although, it will not be possible to quantitatively account and reduce the data for the difference in CG position. Therefore, this will be taken into account in the conclusions.

2.3.3 Test results, analysis and discussion

The stall points were identified in the FTI data by combining event markers (set by the FTE during flight) with four primary indicators, which:

1. Pitch-break: sudden increase in nose-down pitch rate and pitch angle, followed by altitude loss.
2. Roll-break: abrupt increase in roll rate and bank angle, leading to a lateral upset and altitude loss.
3. Sudden altitude loss with g-drop: rapid reduction of vertical acceleration below 1 g, which consistently preceded or accompanied pitch- and roll-breaks.
4. Elevator stroke limitation – full aft stick displacement without additional effect on aircraft attitude.

The stall point was taken as the first occurrence of any of these phenomena. Entry speed was identified at 1.1 Vs, and the line traced between entry and stall speed determined the actual deceleration rate. The associated altitude loss was then extracted from the altitude trace.

All stall points were post-processed to ensure compliance with certification criteria, specifically those defined in ASTM F2245 and CS-23 §23.201/§23.203. Acceptable stalls were further used in the regression analysis (stall speed vs entry rate) to interpolate stall speed at the reference 1 kn/s deceleration rate required for certification.

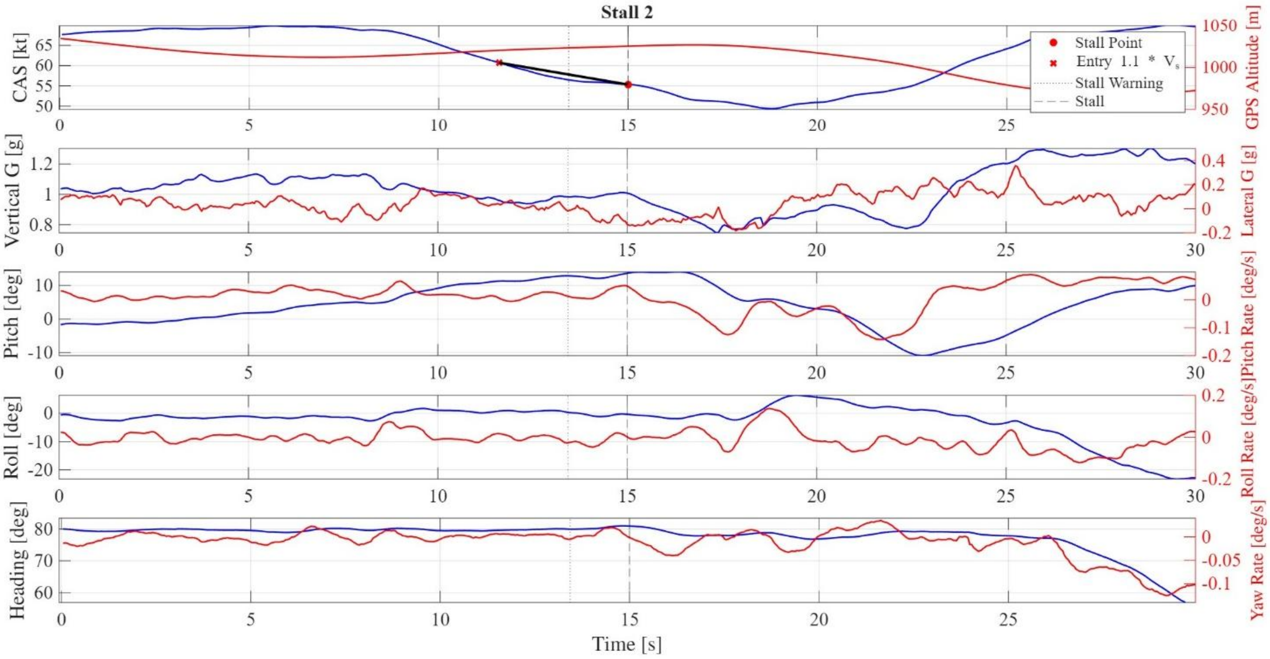


FIGURE 7: SW128 STALL TEST POINT.

Figure 7 is showing one of the SW128 stalls, identified by a pitch-break with consequent drop in altitude and g-drop.

The reduction method for the stall speed is based on the mass of the aircraft at each stall point. The procedure is explained and outlined in Appendix B. In the following table there are summarized and gathered the data points results, reduced for their relative mass. All the plots and analysis of the stall points for both

	HDG [°]	ER [kn/s]	CAS std [kn]	Warn. [kn]	ALT loss [ft]	Stall Type	PIC force	Certification
SW121	90	0.68	47.6	48.1	243	Pitch-break	Positive	Acceptable
	90	0.32	48.0	49.1	171	Stroke-lim.	Positive	Acceptable
	90	1.01	45.4	46.3	128	Pitch-break	Positive	Acceptable
	270	0.69	48.0	49.9	167	Stroke-lim.	Positive	Acceptable
	270	0.97	46.6	47.5	233	Pitch-break	Positive	Acceptable
	270	1.80	44.6	47.1	171	Pitch-break	Positive	Acceptable
SW128	77	0.86	55.6	57.0	168	Pitch-break	Positive	Acceptable
	80	1.58	55.5	59.8	188	Pitch-break	Positive	Acceptable
	82	1.59	55.1	58.4	110	Pitch-break	Positive	Acceptable
	260	1.74	54.1	55.7	198	Pitch-break	Positive	Acceptable
	260	0.91	56.7	58.0	189	Roll-break	Positive	Not Acceptable
	260	1.77	53.3	54.3	189	Pitch-break	Positive	Acceptable

aircrafts are gathered in Appendix C.

TABLE 17: STALL TEST RESULTS.

An analysis against certification criteria for both aircrafts can be performed:

- **SW121** – All six test points were declared acceptable. The stall break was consistently accompanied by either a pitch-down or control stroke limit, both in compliance with certification requirements. Stick-force remained positive throughout. Altitude losses were moderate (128–243 ft) and within acceptable margins for this weight class. The stall warning showed consistency with the manoeuvre dynamics, all the stall points were preceded by stall warning. The stall warning was both audible and visual, making it compliant for the certification.

- **SW128** – Out of six test points, one was not acceptable: The stall at 56.7 kn CAS (ER = 0.91 kn/s) was characterised by a roll-break, which is not considered acceptable from a certification standpoint. The remaining points showed altitude losses between 110–198 ft, consistent with expectations. The stall warning showed consistency with the manoeuvre dynamics, all the stall points were preceded by stall warning. The stall warning was both audible and visual, making it compliant for the certification.

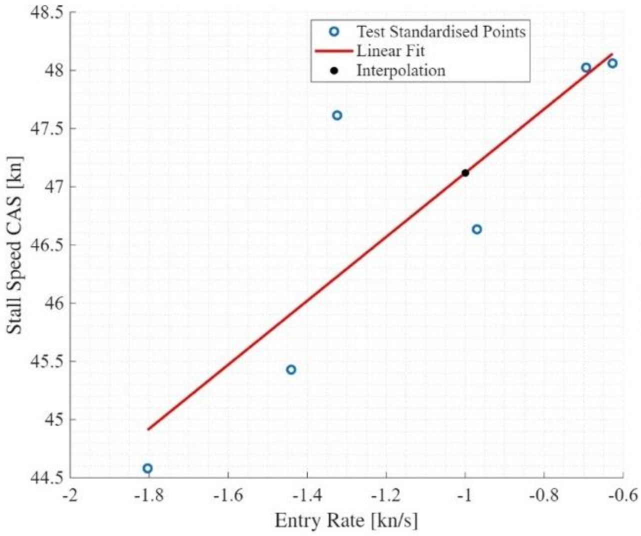


Figure 8: SW121 Stall Speed Interpolation.

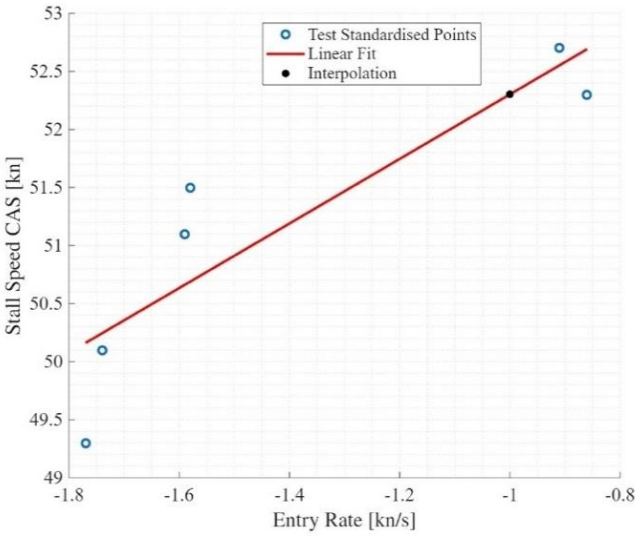


Figure 9: SW121 Stall Speed Interpolation.

The final regression fits of the mass-standardised data are presented in Figures 8 and 9. Interpolation at the reference entry rate of 1 kn/s gives Vs* = 47.1 kn for the SW121 and Vs* = 52.3 kn for the SW128. Certification requires stall speed to be defined at 1 kn/s. Since achieving this rate precisely in flight is impractical, stall speed is regressed against entry rate and interpolated at the reference value. Figures 8 and 9 show the regression curves for the two aircraft.

- SW121: The interpolated stall speed of 47.1 kn is about −2.9 kn lower than the POH value of 50 kn CAS. The difference is within normal scatter and may reflect a slightly aft CG. Overall, results are consistent with expectations.
- SW128: The interpolated stall speed of 52.3 kn is about +1.3 kn above the POH value of 51 kn CAS. The systematic nature of this difference suggests test-condition influences, such as:
  - aft CG location, which increases stall speed.
  - turbulence, potentially biasing stall recognition by inducing earlier g-drop signatures.
  - sensitivity of the electric propulsion system to rapid power changes.

In summary, both aircraft results align broadly with POH references. Minor deviations are attributable to test conditions, and the regression method proved effective for standardising data to certification requirements.

Error Sources and Data Quality

The quality of the stall test data was influenced by instrumentation, environment, and procedural limits. Stalls are short but require high precision in timing and IAS, as certification margins are tight (±1 kn). Turbulence was the main contributor to scatter, occasionally altering attitude, inducing premature g-drops, or producing atypical signatures (e.g. higher stall speed at higher entry rates). Multiple points were re-flown, but repeatability remained limited, particularly for the SW128, whose endurance restricted retries.

Aircraft factors also played a role: neither aircraft was tested at MTOM with forward CG. Mass was corrected via regression, but the aft CG offset could not be and must be considered against POH/regulatory values. For SW121, fuel burn required post-flight mass/CG reconstruction. For SW128, electric propulsion sensitivity to throttle reductions may have affected deceleration.
Data consistency was acceptable, with good FTC–FTI correlation and stall warnings preceding all valid points, in line with CS-LSA/ASTM. One SW128 point was rejected for roll-break, highlighting the residual role of human judgement in stall type classification.
In summary, turbulence and CG deviations were the main sources of error, with secondary contributions from event marking and propulsion dynamics. These explain the observed ±1–2 kn scatter and occasional deviations from theory. Data quality remained sufficient for regression and certification comparison, within the noted limitations.

2.3.4 Conclusion and recommendations

The primary objective of the stall testing campaign was to evaluate the stall behaviour of the SW121 and SW128 in clean configuration (flaps 0) and verify compliance with ASTM F2245 and CS-LSA. A secondary goal was to compare their aerodynamic characteristics.
Both aircraft showed clear, repeatable stall warnings (audible and visual) with sufficient margin before stall. Stick force remained positive up to stall, and roll/yaw controllability was maintained with no input reversal, confirming compliance with controllability and warning requirements.
SW121: All six test points were acceptable. Stall occurred either by pitch break or elevator stroke limit, both certifiable. Altitude losses were moderate and consistent with POH. The interpolated stall speed was 47.1 kn, ≈ −2.9 kn from the POH (50 kn CAS), within scatter and explainable by the slightly aft CG. Overall behaviour was benign and compliant.
SW128: Four of six points were acceptable. One was rejected for roll-breaks, which are not certifiable. The interpolated stall speed was 52.3 kn, about +1.3 kn above POH (51 kn CAS). The small but systematic difference may result from aft CG, turbulence, or sensitivity of the electric propulsion system to throttle changes. Stall warning was satisfactory, but the roll-break might indicate a potential handling issue needing further investigation.
Deviations from certification conditions included operation below MTOM and with CG slightly aft of the forward limit. Mass effects were corrected by regression, but CG differences could not be removed and remain a limitation. Turbulence also reduced repeatability and contributed to scatter.
Despite these constraints, regression proved effective, and results were broadly consistent with POH and regulatory expectations. The SW121 showed fully compliant performance, while the SW128 was generally acceptable with isolated deficiencies requiring follow-up.

Recommendations

- **R3-01**: Repeat stall tests for both aircraft at MTOM and with the CG at the most forward permissible position, to fully meet certification standards.
- **R3-02**: For the SW128, conduct dedicated investigations into roll-break stall modes, including tests in smoother atmospheric conditions and expanded CG ranges.
- **R3-03**: Improve wind determination methods for all stall campaigns (e.g. standardise pre-flight cloverleaf manoeuvres), to reduce turbulence bias and improve data quality.
- **R3-04**: Consider implementing automated stall event markers linked to g-drop detection in the FTI, to minimise human reaction time error in stall identification.
- **R3-05**: Assess the impact of propulsion system dynamics on deceleration rates and stall entry characteristics.

2.4 Landing Performance

The objective of the landing performance test is to determine the minimum landing distance (to full stop) of the aircraft and to verify its compliance with EASA CS-LSA [5] (CS-23 §23.75 [8]) and ASTM F2245 (§4.4.4) [4] requirements. The test results are also compared against the performance data declared in the POH of the

SW121 and SW128. Certification criteria require that the landing manoeuvre be conducted from a stabilised approach, with the aircraft crossing the 50 ft reference obstacle at a calibrated airspeed not less than 1.3·Vs0. For both the SW121 and the SW128, this corresponds to an approach speed of approximately 60 KIAS. The landing must be completed with proper alignment to the runway centreline, and the flare and braking phases must be smooth, without loss of directional control. The aim of the campaign is therefore twofold:

1. Establish the minimum landing distances of both aircraft under similar environmental conditions.
2. To confirm compliance with the regulatory and POH requirements, while documenting any deviations due to test constraints (e.g. turbulence, runway surface, mass variations).

2.4.1 Test Method and Conditions

The following sequence was applied to each landing test point:

1. Altimeter setting: Prior to the approach, the altimeter was set to QFE, ensuring that the 50 ft reference could be directly monitored from the instrument.
2. Stabilised approach: The pilot stabilised the aircraft in the prescribed configuration and speed.
3. Condition log: The FTE recorded the relevant environmental and aircraft configuration parameters.
4. 50 ft obstacle: As the aircraft passed 50 ft AGL, the pilot called aloud the indicated airspeed (V_{50}). The FTE simultaneously flagged the event and noted the groundspeed.
5. Touchdown: Upon main wheel contact, the pilot called out touchdown speed. The FTE flagged the event and recorded groundspeed.
6. Braking phase: The pilot applied full brakes to a full stop while maintaining centreline alignment. The FTE recorded the stop event with both marker and chronometer.
7. Final condition log: The FTE logged all the gathered test point and environmental information.

Instrumentation

The landing manoeuvre is highly dynamic, with rapid transitions between phases. To balance accuracy and pilot workload, instrumentation was kept deliberately simple, with redundancy built in through a combination of manual logging and FTI.

Measurement	Flight Card	Data - SW121	Accuracy	Data - SW128	Accuracy
Time	Chronometer	FTI GPS time	± 1 s	FTI GPS time	± 0.01 s
Speed	Displayed GS	FTI GPS speed	± 0.1 m/s	FTI GPS speed	± 0.01 m/s
Engine RPM	Displayed RPM	GEA 24 EIS	± 1 RPM	EPSI570C	± 1 RPM
Engine MAP / PW	Displayed MAP	GEA 24 EIS	± 3.4 Pa	EPSI570C	± 10 W
Height	Altimeter	GSU 25C + GPS FTI	± 1 m	AHRS + GPS FTI	± 1 m
Distance	None	GPS coordinates	± 1 m	FTI GPS coordinates	± 1 m

TABLE 18: LANDING MEASUREMENTS.

Test Conditions

The landing tests were flown at circuit altitude and with approaches stabilised on the extended centreline. Atmospheric parameters were collected prior to each run and confirmed immediately after landing. The key conditions were:

	QNH [hPa]	OAT [°C]	Flaps	TOM [kg]	CG [% MAC]	Headwind [kn]	Runway slope [°]
SW121	1017	23	+2, ½ AB	576.0	28.79	8	negligible
SW128	1017	24	+2	583.4	29.20	7	negligible

TABLE 19: LANDING CONDITIONS.

Two important considerations apply to the test conditions:

1. Certification standards require that landing tests be carried out with the aircraft loaded at the most forward CG position. This criterion could not be met, as the actual test configurations corresponded to 28.79% MAC for the SW121 and 29.5% MAC for the SW128. The deviation was due to practical limitations in achievable loading.
2. The tests were performed in the presence of turbulent winds. While mean values of wind speed and direction were recorded (Table 20), short-term fluctuations during the manoeuvre were not measured. These variations likely affected the measured landing distances and will be taken into account during the analysis.

The wind conditions turn out to be compliant with the proof of compliance of the CS [8].

Reduction Methods

The following correction methods were applied to normalise the landing performance results:

- Standardisation of measured values to the POH reference mass of 600 kg.
- Application of wind correction factors, as prescribed in the POH and [13].
- Standardisation to reference atmospheric conditions (pressure altitude and OAT).

The detailed formulae used for each correction, as well as those applied for POH value adjustments, are presented in Appendix B.

2.4.2 Test results, analysis and discussion

Two approaches were used to analyse the landing test: flight test cards and reduced data from the on-board FTI system. The card-based method applied a simplified kinematic model, dividing the manoeuvre into descent and ground roll phases, each approximated as uniformly accelerated motion (Appendix B). The FTI-based method served as the reference, offering higher accuracy and avoiding manual logging errors. It relied on aircraft attitude, air data (IAS, pressure altitude, OAT), and GPS-derived quantities (latitude, longitude, groundspeed, altitude) recorded throughout the manoeuvre.

Data Processing Procedure

The FTI data were processed according to the following steps:

1. The manoeuvre start was defined at the instant the aircraft crossed 50 ft above runway elevation, identified from the altitude vector.
2. During descent, pitch attitude remained nearly constant until the flare, where a gradual pitch increase was observed.
3. Touchdown was identified when GPS altitude equalled runway elevation, confirmed by a step change in the normal acceleration vector. This event, expressed in GPS coordinates and elapsed time, marked the transition from descent to ground roll.
4. The end of the manoeuvre was defined when the aircraft reached a full stop on the runway, corresponding to groundspeed reaching zero.
5. GPS coordinates, airspeed, and elapsed time were extracted for all key points.
6. Ground roll distance and total landing distance (from 50 ft to full stop) were computed from the recorded GPS track.

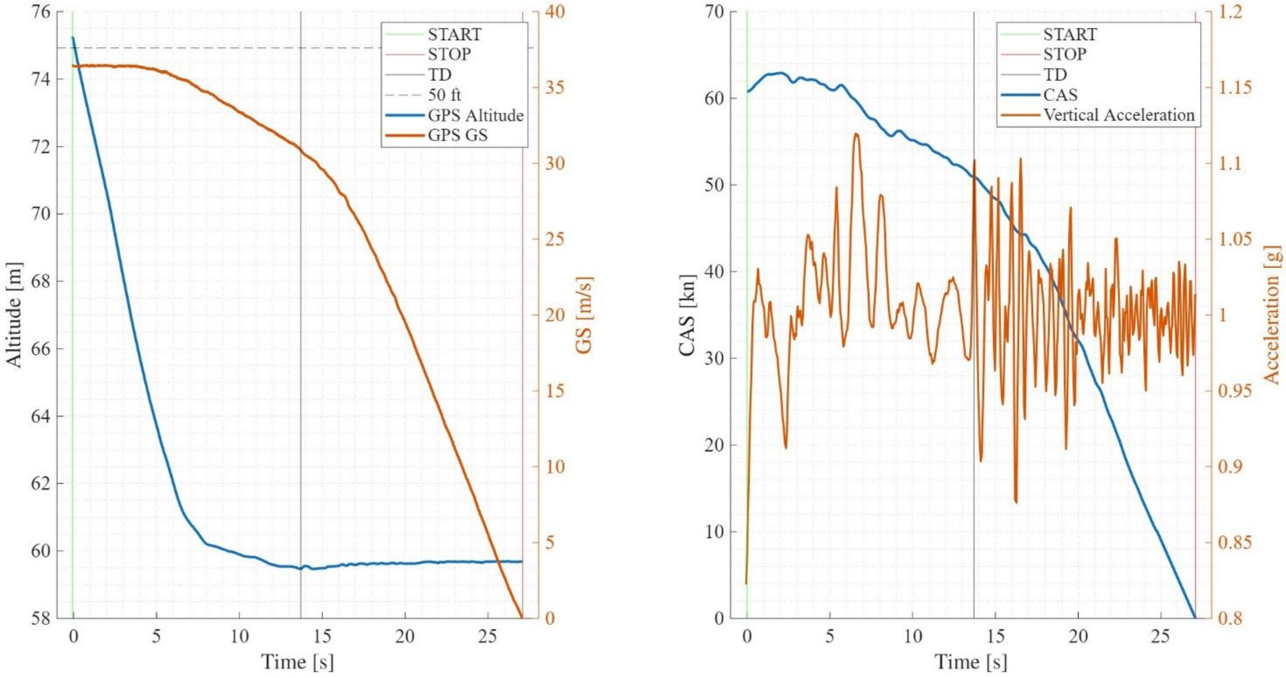


FIGURE 10: SW128 LANDING MANOEUVRE.

Figure 10 illustrate the full landing sequence of the Velis Electro SW128. Event markers for touchdown were found to align consistently with the observed aircraft dynamics. The plots also highlight the identification of key phases and the corresponding extraction of airspeed, altitude, and distance values.
The following table summarises the raw parameters obtained from the analysis for both aircraft types.

	CAS [kn]	SW121	SW128
FTI Data	50 ft	68.0	60.8
	TD	64.2	53.7
	STOP	0	0
FTC Data	50 ft	68.0	62.0
	TD	63.0	53.0
	STOP	0	0
Certification	50 ft (1.3*Vs0)	62.4	58.5

TABLE 20: LANDING RAW DATA.

From Table 21 shows that the manoeuvre for both aircrafts was performed in accordance with the specifics of the certifications. Although, we can also see that, probably in order to maintain a certain safety margin in turbulent tailwind conditions, both pilots were quite conservative with the approach speeds. This might have affected the final results, most probably dilating the landing distances.
Variations in instantaneous wind speed may also have contributed to results deviations. Such deviations, while minor, could influence both the ground roll and the total landing distance, tending to increase them slightly.
The agreement between FTI and FTC data further confirms the reliability of the manual logging process and demonstrates that the Flight Test Engineers performed the task with accuracy.

Landing Distance

The table below presents the corrected landing distances. Both FTI and FTC data were reduced for the specific test conditions, notably wind, mass and air density, using the methodology described in Appendix B. The POH values were interpolated to account for the runway elevation and OAT.

	FTI corrected data		FTC corrected data		POH data	
Distance [m]	**Ground roll**	**50 ft**	**Ground roll**	**50 ft**	**Ground roll**	**50 ft**
SW121	340	515	301	502	326	511
SW128	351	547	321	551	202	602

TABLE 21: CORRECTED LANDING RESULTS.

The results show that the measured landing distances generally exceeded the POH reference values. For the SW121, the corrected 50 ft landing distance agreed closely with the POH (515 m vs 511 m), while the ground roll was about 4% longer. For the SW128, larger discrepancies were observed: the corrected 50 ft landing distance was within 9% of POH, but the ground roll exceeded the POH by more than 70%. Despite this, both aircraft were flown in compliance with the certification approach speeds, and the results remain representative of the expected performance envelope. The most significant deviations are attributed to the ground roll of the SW128. Contributing factors likely include:

- **Pilot technique:** Higher-than-required approach speeds were flown to maintain safety margins in turbulent tailwind conditions, which tends to increase landing distance.
- **Wind variability:** Instantaneous gusts not captured in average wind corrections may have biased the results, especially during the flare and rollout.
- **Runway conditions:** Surface friction and slope may differ from the idealised assumptions underlying POH data.

The close agreement between FTI and FTC results demonstrates the reliability of both methods and validates the manual logging process. Overall, the tests were conservative but consistent with certification requirements. The results can therefore be considered technically valid, though for certification purposes, repeated compliant runs would be necessary to reduce scatter and confirm the observed trends.

Error Sources and Data Quality

The accuracy of the landing distance data is influenced by both the instrumentation employed and the dynamic nature of the manoeuvre. Manual entries on the flight test cards, such as time and speed calls, are inherently less precise than the digital outputs of the FTI. For example, timekeeping with a chronometer has an accuracy of about ±1 s, whereas GPS-based timing provides a resolution of ±0.01 s. Considering that the ground roll typically lasts 15-20 s, a one-second error in timing could correspond to uncertainties of ±20–25 m in the estimated distance.

Distance, determined from GPS track coordinates, is generally reliable within ±1 m per point. However, integration of small position errors over the entire landing roll can accumulate. Altitude measurements around the 50 ft criterion have a resolution of about ±1 m, which translates into an uncertainty of ≈10 m in the corresponding landing distance reference.

In addition to instrumentation limits, other external and operational factors can affect the results:

- **Turbulent wind conditions:** The recorded mean values cannot capture instantaneous gusts, which may have influenced flare timing and rollout distances.
- **Pilot technique:** Conservative approach speeds and variations in flare execution directly influence ground roll and braking effectiveness.
- **Mass and CG position:** Certification requires MTOM and the most forward CG, but the actual tests were performed at slightly aft CGs; this may reduce rotation effort but alter braking dynamics.
- **Runway surface:** Friction coefficient and slope deviations from the POH assumptions contribute to scatter.

Overall, the combined effect of instrumentation accuracy, environmental variability, and human factors can account for variations of ±25–35 m in the ground roll and ±35–45 m in the total distance from 50 ft. Nevertheless, the strong agreement observed between FTI and FTC results indicates that data quality is sufficient for the intended performance assessment, with deviations remaining within acceptable margins for compliance evaluation.

2.4.3 Conclusion and recommendations

The primary objective of determining the minimum landing distance to full stop was met, with both aircraft completing the manoeuvre in accordance with CS-23/LSA and ASTM requirements. The prescribed approach speed of approximately 1.3·Vs0 was achieved, with the pilots opting for slightly conservative margins in turbulent conditions. Proper stabilisation on final and smooth execution of flare and braking phases ensured that the certification criteria for approach, obstacle clearance, and directional control were satisfied.

After correction, the SW121 exhibited agreement with the POH, with the 50 ft landing distance within 1% and the ground roll exceeding the handbook by only about 4%. The SW128, in contrast, showed a larger deviation: the 50 ft distance was within 9% of POH, but the ground roll exceeded the published value by more than 70%. These discrepancies are primarily attributed to pilot conservatism in approach speeds, instantaneous wind variations not captured in the average corrections, and potential runway surface effects. Compliance with certification requirements was therefore demonstrated, although the required MTOM condition could not be achieved due to inability to perform the flight for only the landing performance testing. This represents a limitation with respect to full compliance but is expected to have only minor influence on the observed results. The turbulence encountered during the tests likewise introduced variability into the dataset, particularly affecting rollout distances. It is worth noting that in order for the aircrafts to be certified for the landing distances, at least 6 attempts, compliant with the regulations, must be made. The multiple data would smooth uncertainties and lead to more accurate estimates.

The consistency between FTI and FTC results confirms that the instrumentation and manual logging methods were adequate for the scope of the campaign, with data quality sufficient for reliable performance assessment.

Recommendations

- **R4-01:** Implement real-time wind measurement at the runway threshold to improve the fidelity of environmental corrections.
- **R4-02:** Ensure test loading permits operation at the MTOM, to achieve full compliance with certification requirements.
- **R4-03:** Conduct a statistically significant number of compliant runs (minimum of six) under comparable conditions, as required for certification evidence.
- **R4-04:** Provide additional pilot briefing on approach speed discipline and braking technique, in order to minimise variability caused by conservative or inconsistent flare execution.
- **R4-05:** Consider supplementary runway surface monitoring to characterise braking friction and slope, reducing uncertainty in ground roll comparisons.

3 Additional analysis - Simulated Longitudinal Stability

The additional analysis focused on investigating the connection between numerical modelling tools, such as *Flow5*, and real-life flight test results. The underlying hypothesis was that *Flow5* could be used to model the complete aircraft configurations (SW121 and SW128) through a triangular panel method. The software allows modelling of all major lifting surfaces, including the wing, horizontal and vertical tails, as well as the fuselage, represented using seven cross-sections. Studying this connection provides interesting and useful results, it allows to make a fast and un-expensive estimate of the aircraft dynamic response. This link, if correctly validated could be an extremely useful tool to implement in aircraft design.

3.1 Test Method and Conditions

Most of the data required for the modelling were retrieved from the POHs of the aircraft and from technical documentation provided by Pipistrel. In cases where specific aerodynamic or inertial properties were not available, approximations were introduced, making use of the graphical tool *ImageJ* and other engineering estimates. It should be noted that *Flow5* only supports control-fixed analyses, consequently, the comparison with experimental data was performed under stick-fixed conditions.

Limitations

It is necessary to mention the limitations of the model that could lead to possible sources of error. Some hypotheses have been made due to partial lack of information:

- The model does not account for propeller interaction. This effect could be significant, as the aerodynamic loads on the blades vary strongly with angle of attack and consequently the response of the aircraft.
- Turbulence and wind gusts are inherently absent in the model. Therefore, direct comparison between test and simulated results might pose some interpretation challenges.
- The exact airfoil of the wing was not publicly available. For this reason, the *FX 63-137* (common airfoil among GA aircrafts and motor gliders) was used for the wing, and a NACA 0009 profile was selected for the tail surfaces.

Conditions

The *Flow5* model analysis have been carried out with the following conditions, regarding the simulation environment and methods:

	PA [ft]	OAT [°C]	CAS trim [kn]	Mass [kg]	CG [% MAC]
SW121	4200	14.5	80	579.8	28.75
SW128	3000	18	77	583.4	29.20

TABLE 22: MODEL CONDITIONS.

The analysis was carried out using triangular panels with uniform density distribution. The stream speed was chosen based on the convergence speed of the oscillations observed during the flight tests, ensuring consistency between test and simulation.

3.2 Test results, analysis and discussion

Figure 11 and Figure 12 below present the aircraft geometry model together with a direct comparison between simulation and flight test results for the SW128. The datasets were aligned in the time domain to enable a rapid and consistent comparison between numerical and experimental results. Specifically, the stick release instant and oscillation amplitude were matched to the first valley of the simulated response. In addition, the forced excitation applied during the test was reproduced in the software to ensure the same input conditions.

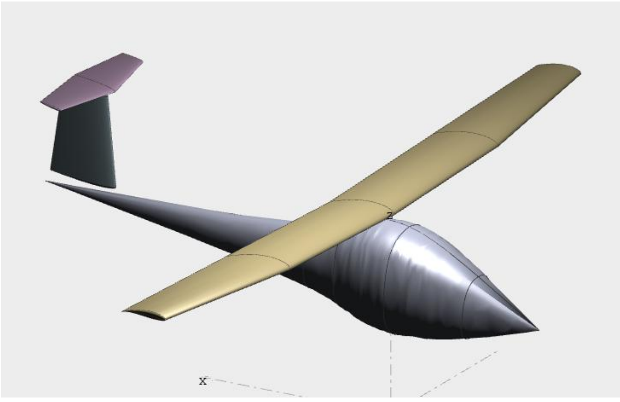


FIGURE 11: FLOW5 MODEL.

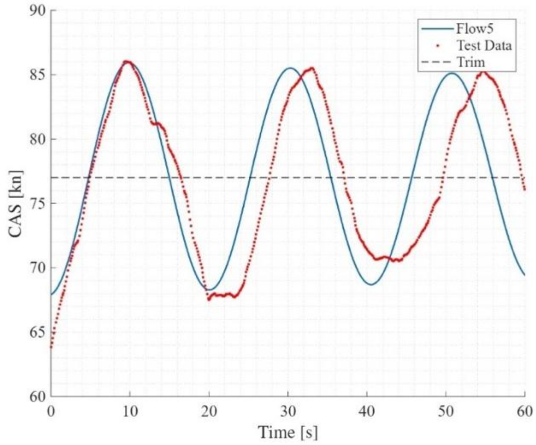


FIGURE 12: FLOW5 COMPARISON.

The averaged stick-fixed results obtained from the flight tests were compared with those predicted by *Flow5*:

	Flight Testing		Flow5			
	T [s]	**ζ [-]**	**T [s]**	**ζ [-]**	**Error T [%]**	**Error ζ [%]**
SW121	24.9	0.00576	21.3	0.00600	- 14.5	+ 4.17
SW128	25.4	0.00654	22.6	0.00700	- 11.0	+ 7.03

TABLE 23: FLOW5 RESULTS.

The oscillation period was systematically underestimated by Flow5 (11–15 %). This can be partially attributed to approximations in inertia properties: Flow5 estimates inertia from geometry, whereas in reality inertia is strongly dependent on internal mass distribution.
The damping ratio was slightly overestimated (4–7 %). This effect is consistent with the absence of propeller dynamics and turbulence in the model, both of which contribute to increased damping in real flight. Despite these limitations, the agreement between simulation and experimental results is considered satisfactory. The deviations fall within the expected range for a simplified potential-flow model.

3.3 Conclusion and recommendations

The comparison between Flow5 predictions and flight test data demonstrates that:

- Flow5 provides a reasonably accurate estimate of the aircraft's dynamic response in stick-fixed conditions.
- The tool is particularly useful for capturing qualitative trends, such as the influence of configuration changes (e.g., SW121 vs SW128) on stability.
- Quantitative accuracy is limited by simplified modelling of inertia, absence of propeller effects, and lack of turbulence representation.

Recommendations

- **R5-01:** Refined inertia modelling: Incorporate more accurate mass distribution data from weight-and-balance documentation or CAD models to improve accuracy of inertial properties.
- **R5-02:** Propeller modelling: Couple Flow5 results with propeller aerodynamic models to capture the effect of thrust and slipstream on stability.
- **R5-03:** Validation campaign: Perform additional flight test points under varied atmospheric conditions to build a more robust validation dataset.
- **R5-04:** Integration in design: Use Flow5 in preliminary design phases to screen configurations and identify trends.

4 Conclusions and recommendations

The flight test campaign described in this report was overall successful. All scheduled test points were completed, and the FTE checklist, organisation, and techniques proved effective. The FTEs conducted multiple training sessions in the simulator prior to the actual flights, which was an optimal strategy to ensure preparedness for the test environment. The collaboration between PICs and FTEs was effective, and the primary objective of obtaining reliable results was achieved.

The dual data-gathering approach also proved feasible. Recording results on flight test cards was within the FTEs' capabilities, and the workload distribution, which included delegating some tasks to the PICs, was efficient and did not distract or overload the pilots. On the critical side, the quality of some FTC data could be improved. Better timing and greater manoeuvre awareness would reduce FTE workload and improve situational awareness during test points.

One of the most significant external factors affecting the tests was wind, and particularly turbulence. While steady wind components can be corrected with reasonable accuracy during data reduction, turbulence cannot be monitored instantaneously or corrected with the same reliability. As a result, even with robust data reduction methods, the analysis remains affected by scatter introduced by turbulent conditions. This was especially evident in the take-off and landing tests, which are highly sensitive to wind effects. In these cases, turbulence not only altered the measured distances but also increased pilot workload significantly, making it challenging to consistently perform the optimal techniques required to achieve minimum distances.

4.1 Conclusions

Overall, the four flight test campaigns confirmed that both aircraft meet their declared performance and regulatory requirements within the tested conditions. Deviations were small and explainable by environmental or procedural factors, rather than systematic performance deficiencies. The SW121 consistently demonstrated compliant behaviour, while the SW128 showed generally positive results with isolated issues requiring further verification.

Take-off

The minimum take-off distance to 50 ft was successfully measured for both aircraft. Corrected results were in close agreement with POH values: SW121 showed about 4% longer distances, while SW128 was within 1%. Certification speed requirements were met, with rotation near 1.1 Vs1 and 50 ft crossing near 1.3 Vs1. Ground roll distances were systematically longer than POH, likely due to turbulence, pilot technique variations, and runway effects. Instrumentation and data collection methods were judged adequate, though reliance on average wind values and lack of runway friction data were limitations.

Longitudinal Dynamic Stability – Phugoid

Phugoid testing demonstrated that both SW121 and SW128 are dynamically stable at 80 KIAS and flaps 0. Oscillations converged consistently, in line with ASTM F2245 requirements. Turbulence had a stronger influence on damping estimates than on oscillation period, making damping ratios less reliable. CG uncertainties, especially on the SW121, also introduced bias.

FTI was essential for reliable analysis, while FTC data showed higher scatter. The objectives were met, but expanded testing across speeds and CG ranges is recommended for full robustness.

Stall

Stall testing in clean configuration confirmed compliance with ASTM and CS-LSA requirements for both aircraft. Both displayed clear stall warnings and maintained controllability.

For the SW121, all test points were valid; stall speeds matched POH within ~3 kn, attributable to CG effects.

For the SW128, most results were valid, but one case showed a roll-break, which is not certifiable and warrants further study. Stall speeds were within 1–2 kn of POH.
Tests were not performed at MTOM or forward CG; while mass corrections were applied, CG limitations remain. Turbulence also reduced repeatability. Despite this, stall behaviour of both aircraft can be considered acceptable, with the SW121 fully compliant and the SW128 requiring additional targeted verification.

Landing

Landing distances to full stop were successfully measured. Both aircraft were flown in accordance with certification techniques, though pilots applied slightly conservative approach speeds under turbulence. Corrected distances for the SW121 matched POH within 1% (50 ft) and 4% (ground roll). For the SW128, the 50 ft distance was within 9%, but ground roll exceeded POH by >70%. This discrepancy was attributed mainly to turbulence, conservative pilot technique, and runway conditions.
Certification compliance was demonstrated, although MTOM testing was not performed. At least six compliant runs would be required for formal certification averaging. FTI and FTC data agreed well, confirming the adequacy of the instrumentation and manual logging methods.

Additional Analysis

Flow5 modelling provided useful insights:

- Stick-fixed phugoid dynamics were captured qualitatively, and results agreed reasonably with flight tests.
- Quantitative deviations arose from simplified inertia modelling, absence of propeller effects, and lack of turbulence representation.
  Overall, Flow5 proved valuable for preliminary prediction of dynamic response, though not sufficient as a stand-alone certification tool.

4.2 Recommendations

List all recommendations as found in Sections 2 and 3.

4.2.1 Take-off Performance

- **R1-01**: Implement real-time wind measurement at the runway threshold (both direction and speed) to improve the accuracy of performance corrections.
- **R1-02**: Ensure test loading permits operation at the most forward CG position, to achieve full compliance with certification requirements.
- **R1-03**: Conduct multiple compliant runs under similar environmental conditions, in order to build a statistically significant dataset for certification use [13].
- **R1-04**: Provide additional pilot briefing and rehearsal focused on rotation timing, to minimise variability due to pilot technique in the dynamic phase of take-off.
- **R1-05**: Consider supplementary instrumentation (e.g. continuous wind sensors or runway surface monitoring) to reduce uncertainties associated with external conditions.

4.2.2 Longitudinal Dynamic Stability – Phugoid

- **R2-01** Conduct phugoid testing in calmer atmospheric conditions or at times of reduced thermal activity to minimise turbulence-induced damping bias.
- **R2-02** Improve wind measurement prior to and during testing (e.g., gust probes, high-frequency pitot-static data) to allow a more accurate decoupling of atmospheric effects from aircraft dynamics.
- **R2-03** Expand the test campaign to include phugoid manoeuvres across the full CG range and at multiple speeds (from 1.1 Vs up to Vne), to confirm stability throughout the operational envelope.
- **R2-04** Enhance mass and CG tracking on the SW121 by improving the accuracy of fuel flow integration, reducing bias in stability assessment.

- **R2-05** Provide tighter pilot technique standardisation (stick input and release timing) to reduce variability between repeated test points.

4.2.3 Stall Performance

- **R3-01**: Repeat stall tests for both aircraft at MTOM and with the CG at the most forward permissible position, to fully meet certification standards.
- **R3-02**: For the SW128, conduct dedicated investigations into roll-break stall modes, including tests in smoother atmospheric conditions and expanded CG ranges.
- **R3-03**: Improve wind determination methods for all stall campaigns (e.g. standardise pre-flight cloverleaf manoeuvres), to reduce turbulence bias and improve data quality.
- **R3-04**: Consider implementing automated stall event markers linked to g-drop detection in the FTI, to minimise human reaction time error in stall identification.
- **R3-05**: Assess the impact of propulsion system dynamics (particularly for electric aircraft) on deceleration rates and stall entry characteristics.

4.2.4 Landing Performance

- **R4-01:** Implement real-time wind measurement at the runway threshold to improve the fidelity of environmental corrections.
- **R4-02:** Ensure test loading permits operation at the MTOM, to achieve full compliance with certification requirements.
- **R4-03:** Conduct a statistically significant number of compliant runs (minimum of six) under comparable conditions, as required for certification evidence.
- **R4-04:** Provide additional pilot briefing on approach speed discipline and braking technique, in order to minimise variability caused by conservative or inconsistent flare execution.
- **R4-05:** Consider supplementary runway surface monitoring to characterise braking friction and slope, reducing uncertainty in ground roll comparisons.

4.2.4 Additional analysis – Simulated Longitudinal Stability

- **R5-01:** Refined inertia modelling: Incorporate more accurate mass distribution data from weight-and-balance documentation or CAD models to improve accuracy of inertial properties.
- **R5-02:** Propeller modelling: Couple Flow5 results with propeller aerodynamic models to capture the effect of thrust and slipstream on stability.
- **R5-03:** Validation campaign: Perform additional flight test points under varied atmospheric conditions to build a more robust validation dataset.
- **R5-04:** Integration in design: Use Flow5 in preliminary design phases to screen configurations and identify trends.

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Appendix

A Flight cards

The complete flight test cards are provided in separate documents, as including them here would considerably increase the length of the report and reduce its clarity.

B Data reduction

Following a collection of data reduction methods used on the test gathered data in order to obtain standardized and repeatable results.

B.1 - Take-off

FTC Data regression

The total take-off distance is determined under the assumption of constant acceleration between measurement points and is calculated as the sum of the individual segments of the take-off run and climb-out. The computational method used is presented below.

On-Ground Acceleration Calculation, assuming liftoff happens about 2 seconds after rotation:

$$a_R = \frac{V_R}{t_R},\ \ t_{LOF} \approx t_R + 2, \quad S_g = \frac{1}{2} a_R \cdot t_{LOF}^2 = S_1$$

Climb segment calculation:

$$a_{cl} = \frac{V_{50} - V_{LOF}}{t_{50} - t_{LOF}}, \qquad S_2 = \frac{1}{2} a_{cl} \cdot (t_{50} - t_{LOF})^2$$

The total (uncorrected) take-off distance:

$$S_{TO} = S_g + S_{cl}$$

Wind Correction Formula

Following the formula to take into account the wind contribution on take-off performance.

$$S_g \ = \ S_{g_{Test}} * \ (1 \pm V_w/V_{LOF})^{1.85}$$

Where:

S_g – no-wind take-off ground distance.

$S_{g_{Test}}$ – take-off ground distance measured in testing.

V_w – wind velocity component parallel to the strip. + is used for headwind and – for tailwind.

V_{LOF} – groundspeed at lift-off with a known wind velocity.

Runway Slope Correction Formula

Following the formula to take into account the runway slope contribution on take-off performance.

$$S_g = S_{g_{Test}} / \left[1 \pm \left(2 * g * S_{g_{Test}} / V_{LOF}^2\right) * \sin(\beta)\right]$$

Where:

$S_{g_{Test}}$ – ground distance measured during testing.

g – acceleration of gravity.

V_{LOF} – groundspeed at lift-off.

β – angle of the slope in radians. + for upslope and – for downslope.

Weight and Power Correction Formulae

Following the formulae to consider the variation in power, weight and air density compared to standard conditions.

$$\begin{cases} S_{1_{Std}} = S_{1_{Test}} \left(\frac{W_{Std}}{W_{Test}}\right)^{2.6} \left(\frac{\sigma_{Test}}{\sigma_{Std}}\right)^{1.9} \left(\frac{N_{Test}}{N_{Std}}\right)^{0.8} \left(\frac{P_{b_{Test}}}{P_{b_{Std}}}\right)^{0.6} \\ S_{1_{Std}} = S_{1_{Test}} \left(\frac{W_{Std}}{W_{Test}}\right)^{2.6} \left(\frac{\sigma_{Test}}{\sigma_{Std}}\right)^{1.9} \left(\frac{N_{Test}}{N_{Std}}\right)^{0.8} \left(\frac{P_{b_{Test}}}{P_{b_{Std}}}\right)^{0.6} \end{cases}$$

B.2 – Phugoid

For the analysis of the phugoid motion, the period and damping ratio were derived from the calibrated airspeed signal, by identifying the peaks and valleys of the free oscillation.

Amplitudes

The maxima and minima of the velocity response were detected.
For each pair of consecutive maxima (or minima), the logarithmic decrement was computed as:

$$\delta = \ln\left(\frac{A1}{A2}\right)$$

where A1 and A2 represent the amplitudes of two successive extrema.

Damping ratio

For each pair of extrema, the damping ratio was estimated using the relation:

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}$$

The final damping ratio was obtained as the average of all computed values.

Period

The time interval between consecutive maxima (or minima) was measured. The phugoid period was then calculated as the mean value of all such intervals.

B.3 - Stall

Wind Calculation and Correction

Wind correction is derived from the data gathered during a dedicated wind determination manoeuvre, which assumes that both wind direction and magnitude remain constant throughout the four legs of the square pattern. The method applies the Carnot theorem to the geometry of the flight path, combined with the vectorial relationship: $V_{TAS} = V_{\omega} + V_{GS}$

By applying this principle to each leg of the pattern, a solvable system of three equations with three unknowns—true airspeed (TAS), wind speed, and wind direction—can be constructed. The fourth leg offers redundancy, allowing the system to be solved three times using different combinations of three legs. The final wind vector is obtained by averaging these three solutions, which enhances robustness and reduces sensitivity to individual measurement errors. During post-processing, the previously determined wind vector is used to correct GPS-derived ground speed data from the stall manoeuvres.

Weight Correction

To enable consistent comparison of stalling speeds across different aircraft loading conditions, test results are corrected to a standard weight using the following relationship:

$$V_S = V_{S_{Test}} \cdot \left(\frac{W_{std}}{W_{Test}}\right)^{\frac{1}{2}}$$

Where:

V_S - weight-corrected airspeed (CAS)
$V_{S_{Test}}$ - test stalling airspeed (CAS)
W_{std}- standard weight
W_{Test}- testing weight

Airspeed Correction
Finally, airspeed values recorded from the aircraft's airspeed indicator (IAS) are corrected to calibrated airspeed (CAS) using conversion tables provided in the aircraft's Pilot Operating Handbooks (POHs) [7][8], specifically those corresponding to the clean configuration. These corrections ensure fidelity between indicated and actual aerodynamic performance

B.4 - Landing
The final data used for the landing distance assessment is the GPS data acquired from the additional FTI system. However, this data is influenced by several factors and must be corrected and standardised to ensure comparability with other test results. Specifically, the raw data must be adjusted to account for wind conditions, runway slope, aircraft weight, and air density. The procedures described and followed in this section are known as the USN and USAF TPS methods.
Following a simplified procedure to reduce the data gathered from the flight test cards to obtain an approximate, but sufficient precision data on the landing distance. The total landing distance is determined under the assumption of constant acceleration between measurement points and is calculated as the sum of the individual segments of the take-off run and climb-out. The computational method used is presented below. All the speeds presented are groundspeeds from GPS.
Descent segment calculation, considering only the horizontal component of the deceleration:

$$a_{ds} = \frac{V_{50} - V_{TD}}{t_{50} - t_{TD}}, \quad S_3 = \frac{1}{2} a_{ds} \cdot (t_{50} - t_{TD})^2$$

On-Ground Acceleration Calculation

$$a_g = \frac{V_{TD}}{t_{TD} - t_{stop}}, \quad S_g = \frac{1}{2} a_g \cdot \left(t_{TD} - t_{stop}\right)^2 = S_4$$

The total (uncorrected) take-off distance:

$$S_{TO} = S_3 + S_4$$

Slope correction
The runway slope correction is identical to the take-off corresponding correction and described in reference [5].

$$S_{4_{Std}} = \frac{S_{4_{test}}}{\left[1 \pm \left(\frac{2gS_{4_{test}}}{V_{TD}}\right)^2 \cdot \sin(\beta)\right]}$$

Wind correction
The wind correction is performed applying the following formulae from reference [6].

$$\begin{cases} V_{TD} = V_{TD_{Test}} - V_w \\ S_{4_{Std}} = S_{4_{Test}} \left(1 + \frac{V_w}{V_{TD}}\right)^{1.85} \end{cases}$$

Weight and density correction

The difference in weight and air density with respect to the standard conditions can be corrected following the procedure proposed by reference [6].

$$\begin{cases} E_h = \dfrac{V_{50}^2 - V_{TD}^2}{2g} \\ S_{3_{\text{Std}}} = S_{3_{\text{Test}}} \left(\dfrac{W_{\text{Std}}}{W_{\text{Test}}}\right)^{\left(2+\frac{E_h}{E_h+50}\right)} * \left(\dfrac{\sigma_{\text{Test}}}{\sigma_{\text{Std}}}\right)^{\left(\frac{E_h}{E_h+50}\right)} \\ S_{4_{\text{Std}}} = S_{4_{\text{Test}}} \left(\dfrac{W_{\text{Std}}}{W_{\text{Test}}}\right)^2 \left(\dfrac{\sigma_{\text{Test}}}{\sigma_{\text{Std}}}\right) \end{cases}$$

Shaft Power Calculation

SW121: the shaft power needs to be calculated from the Rotax 912 ULS manual through the values of RPM and MAP.

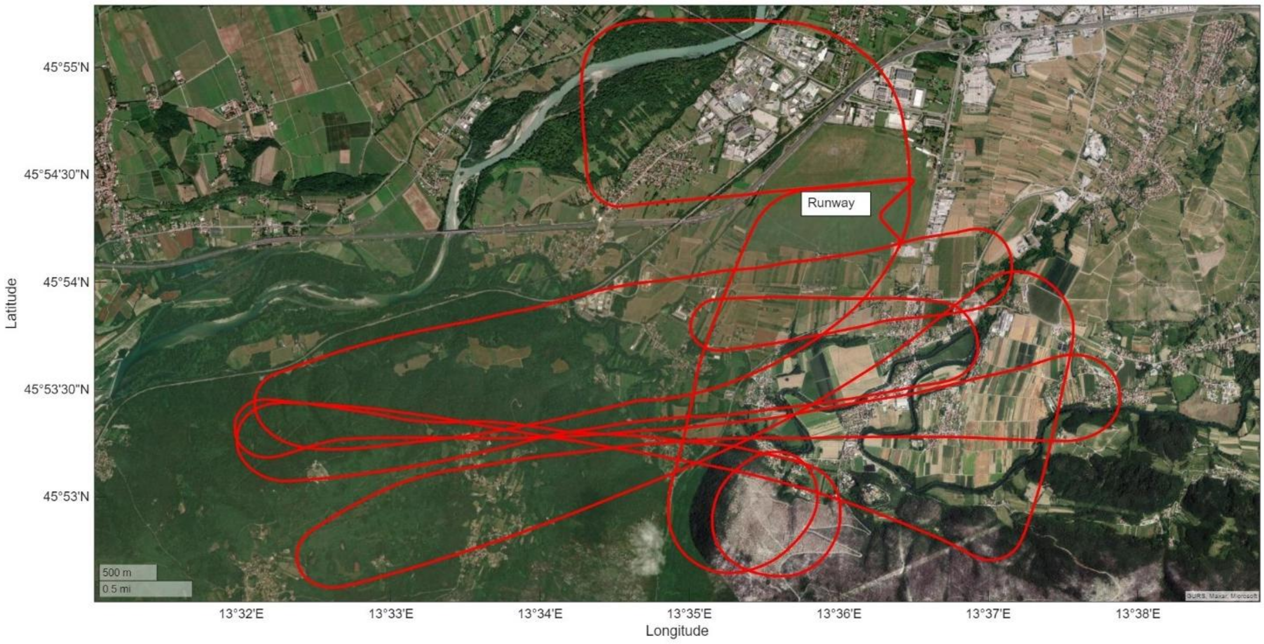
SW128: the shaft power needs to be calculated from the electric power exiting the batteries and engine RPM, knowing the efficiencies of the Electronic Speed Controller (ESC) (η_{esc}), the motor (η_m) and the cables (η_{cbl}). The following formula might be used: $P_b = P_{el} / (\eta_m * \eta_{cbl} * \eta_{esc})$. Said efficiencies can be found in the manufacturer's component manuals.

C Data analysis

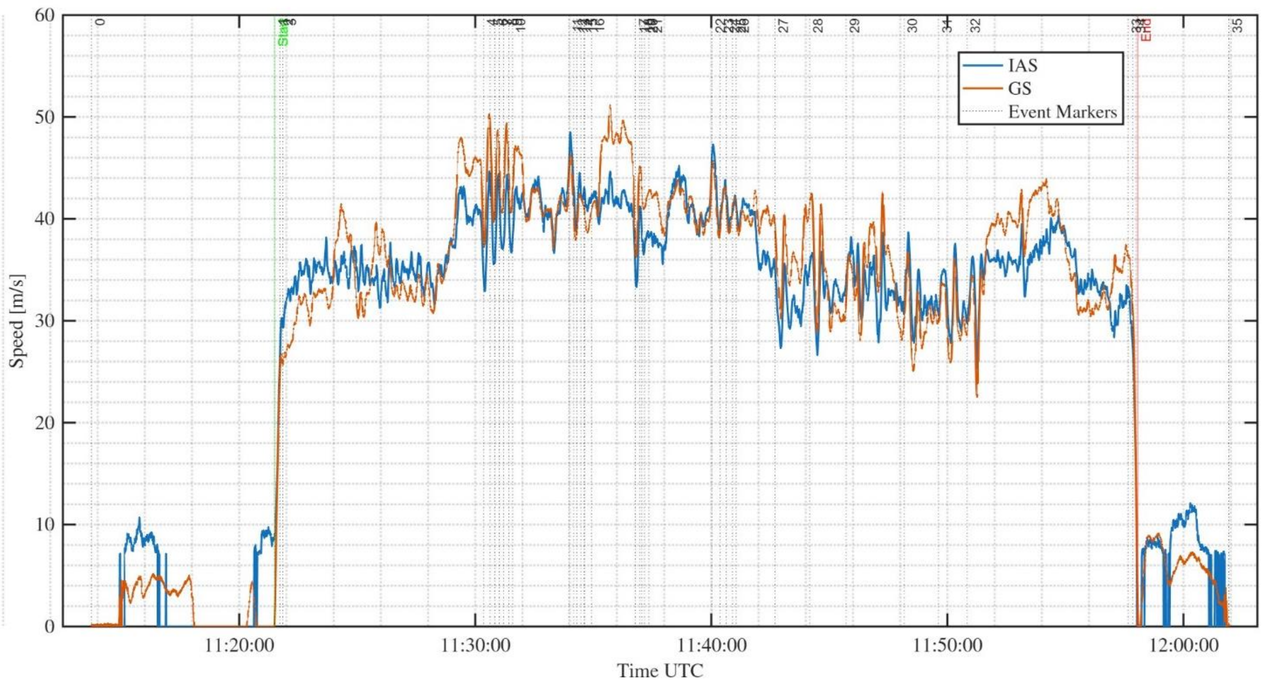
In this section are gathered the additional plots and data used for the analysis and the result discussion.

SW128 – Velis Electro

In the figure is shown the 2D test flight trajectory of the SW128

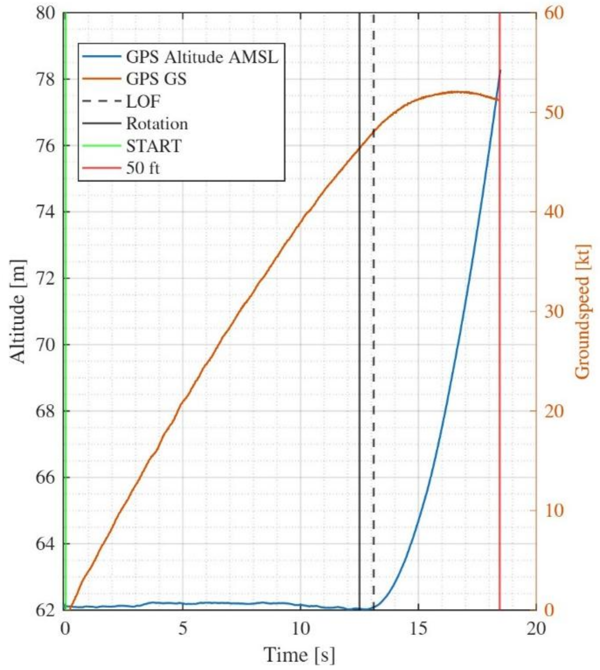


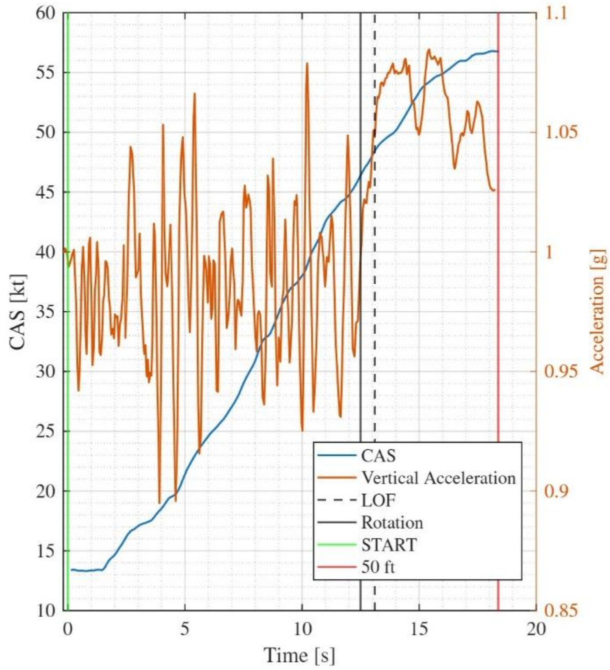
IAS and GS during the test flight mission.



Take-off Performance

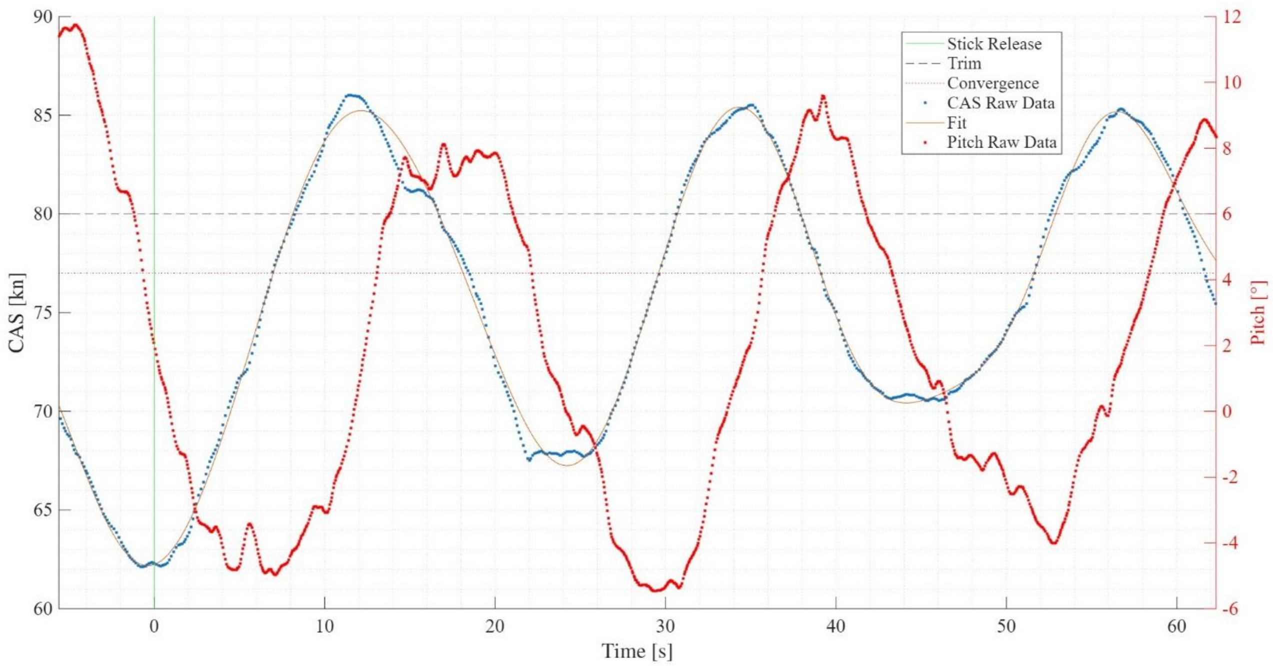
Take-off development of speeds and altitude from FTI data.



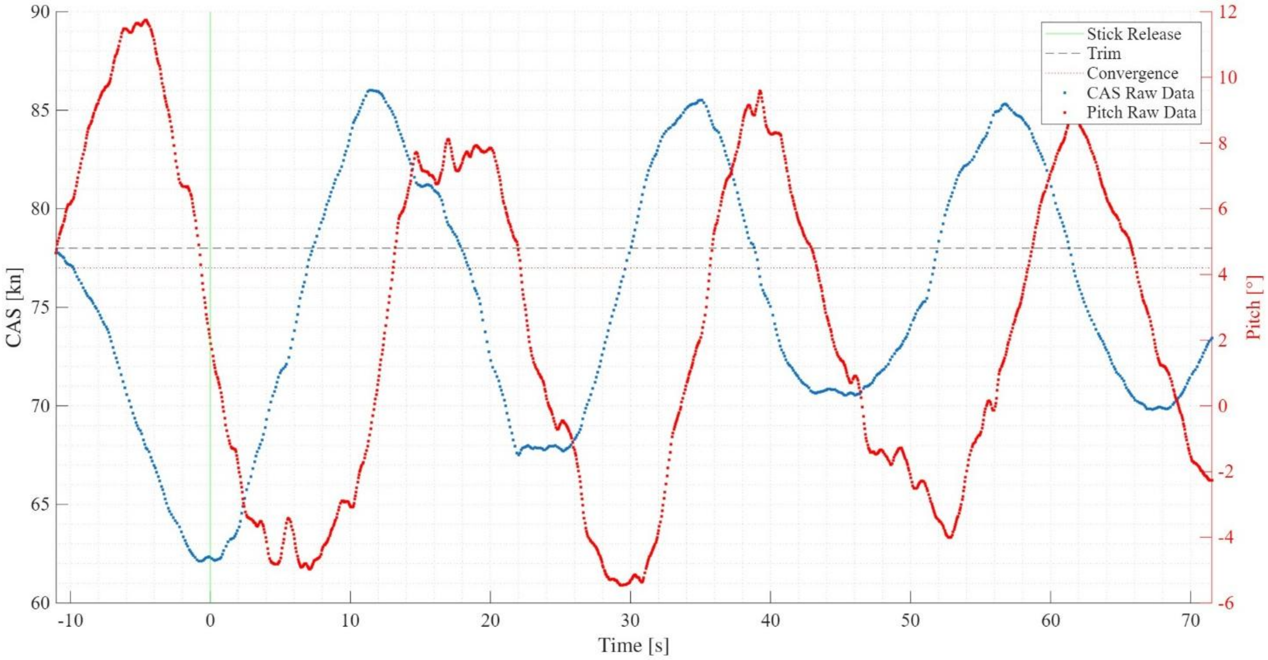


Longitudinal Dynamic Stability – Phugoid

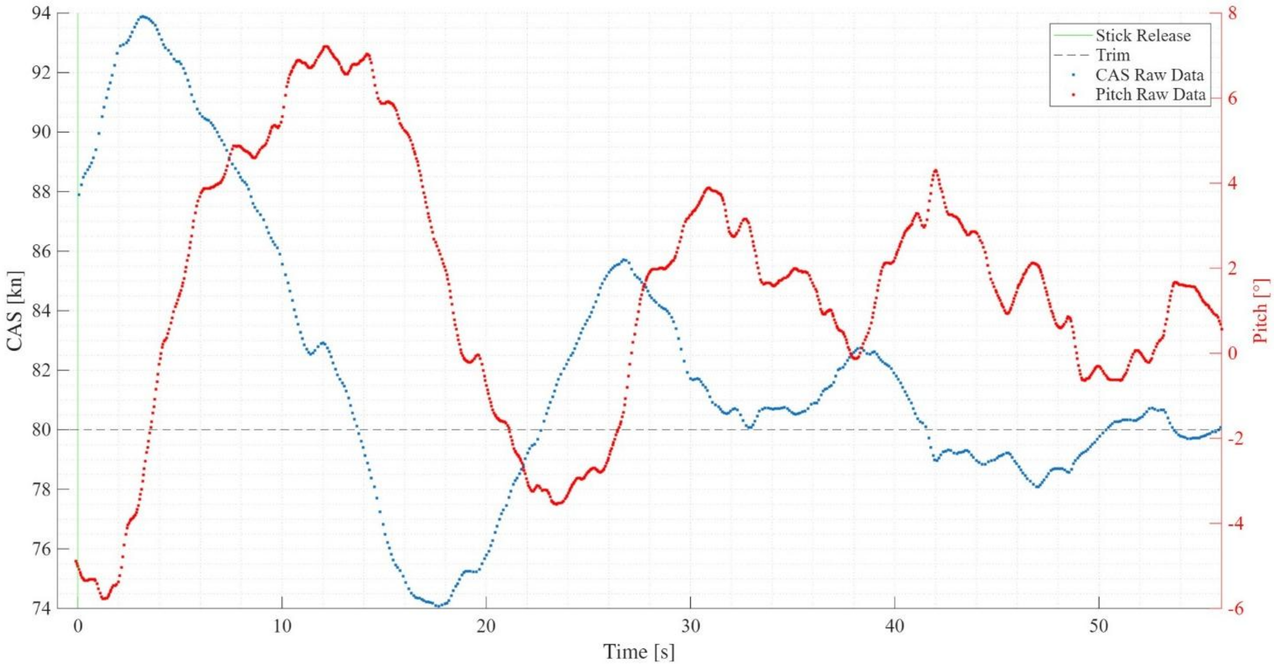
Stick Free test point.



Stick Fixed test point.

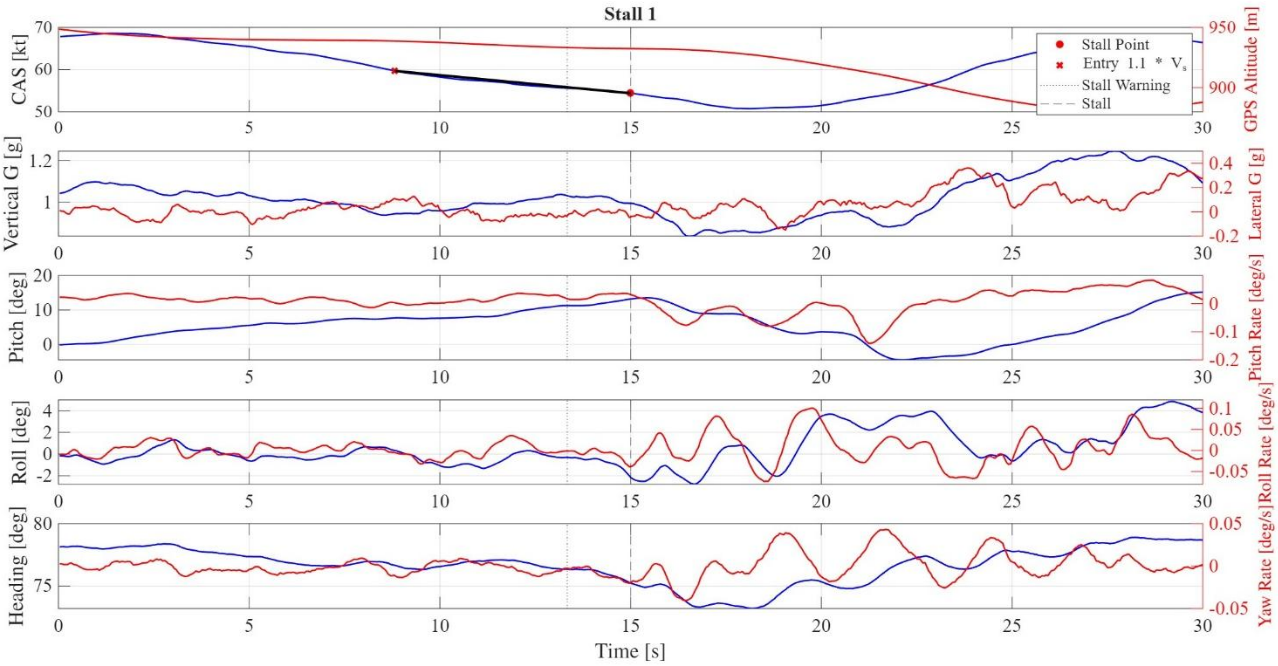


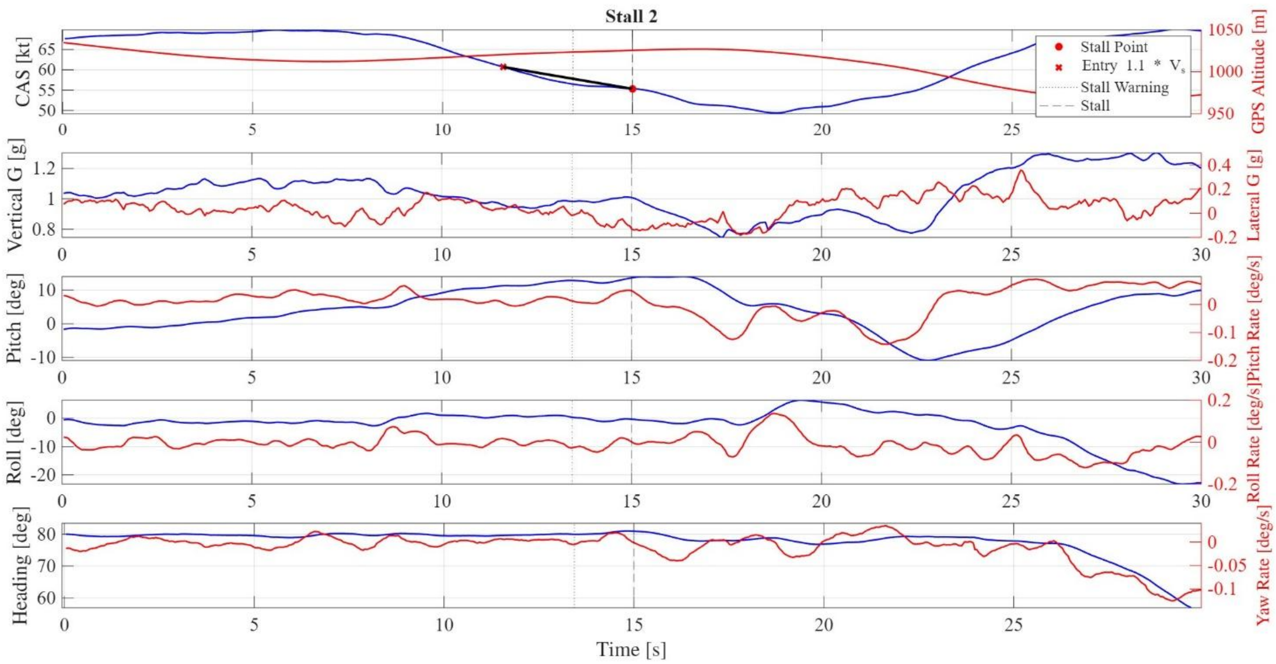
Stick Fixed failed test point example.

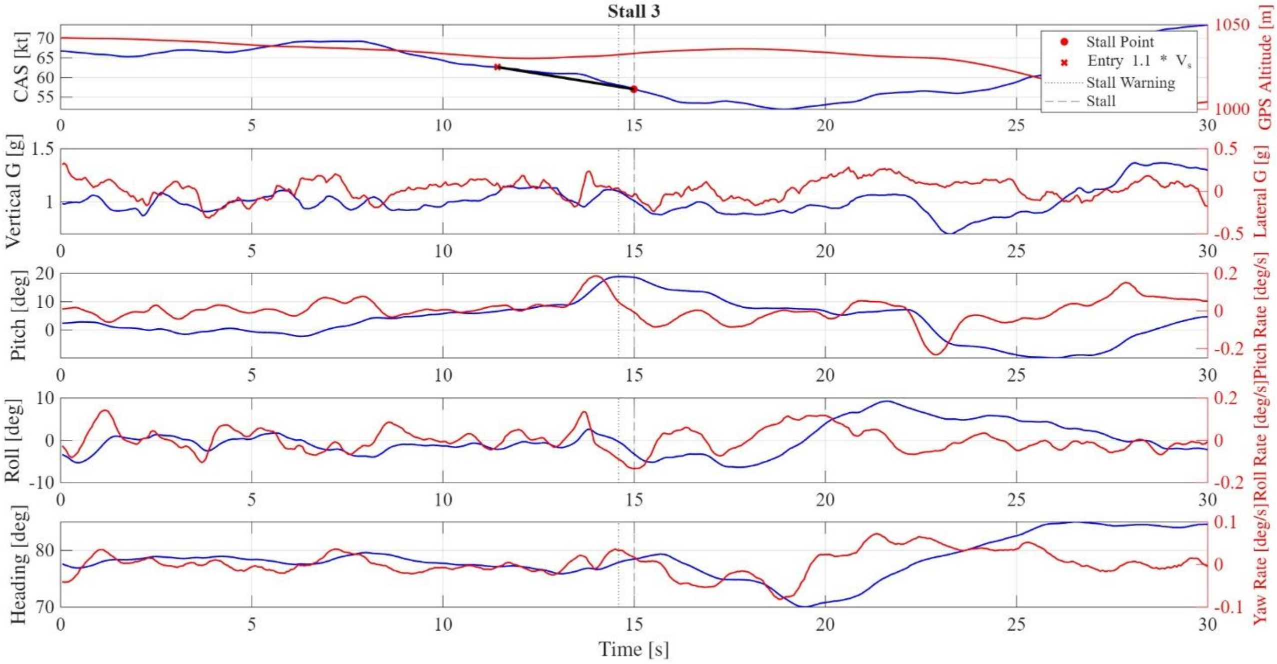


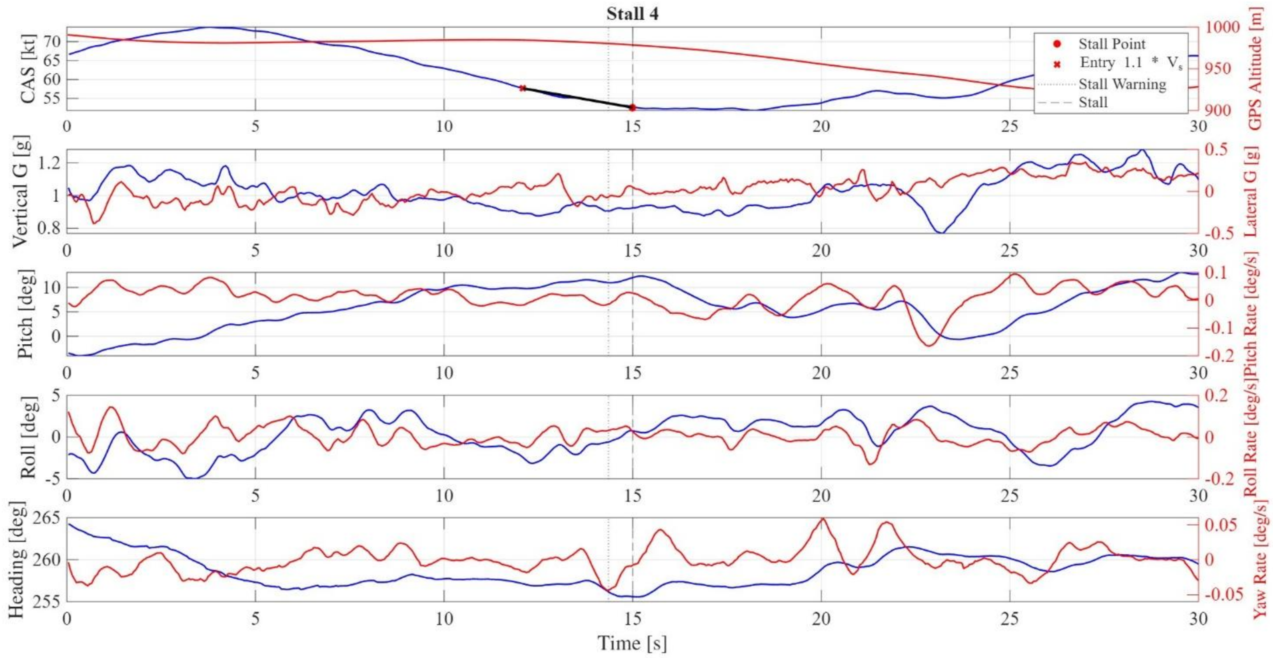
Stall Performance

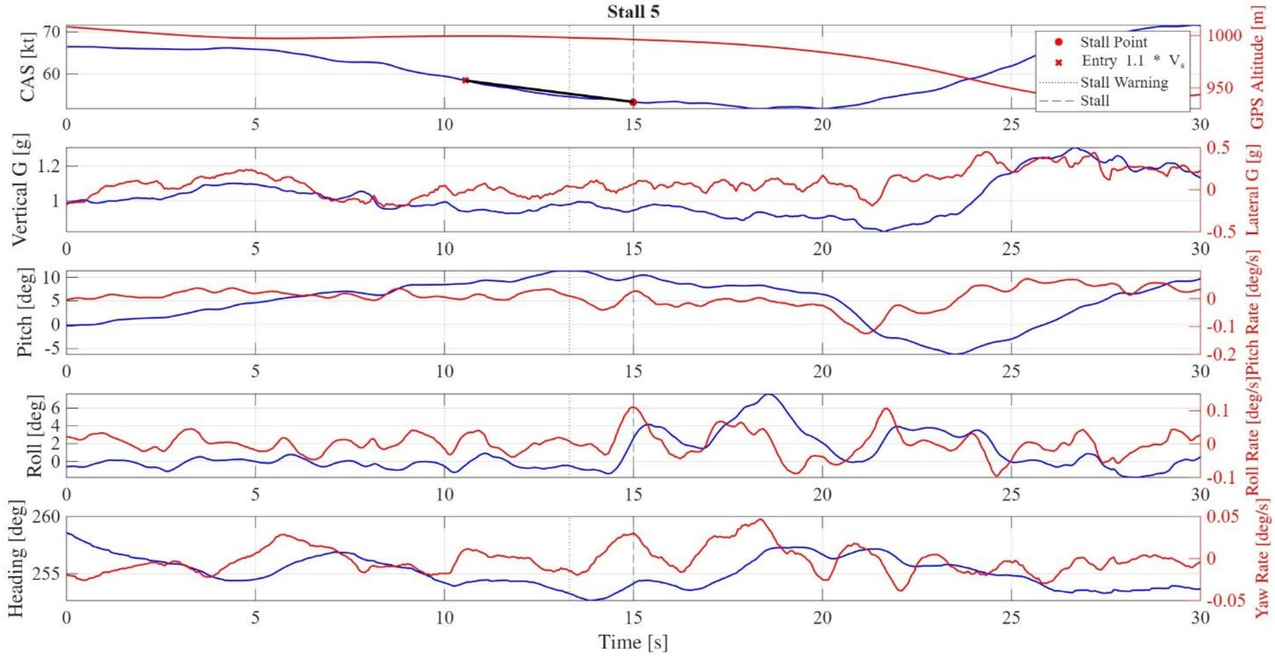
Gathering of all the SW128 stall test points.

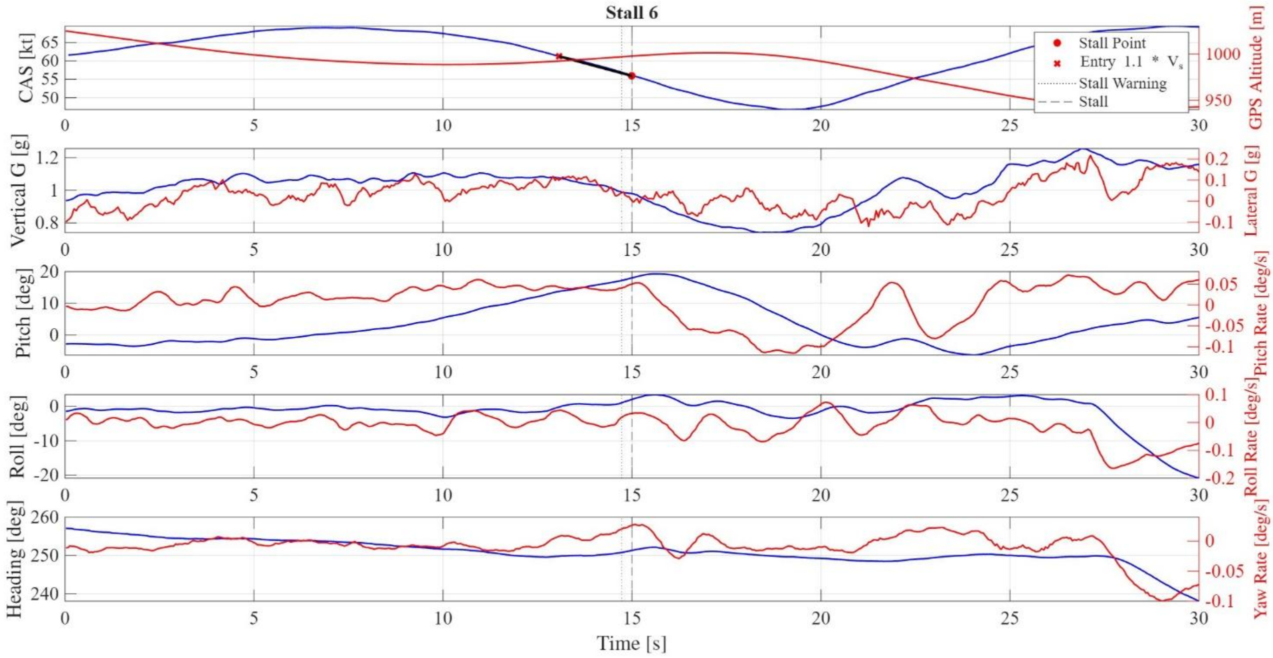




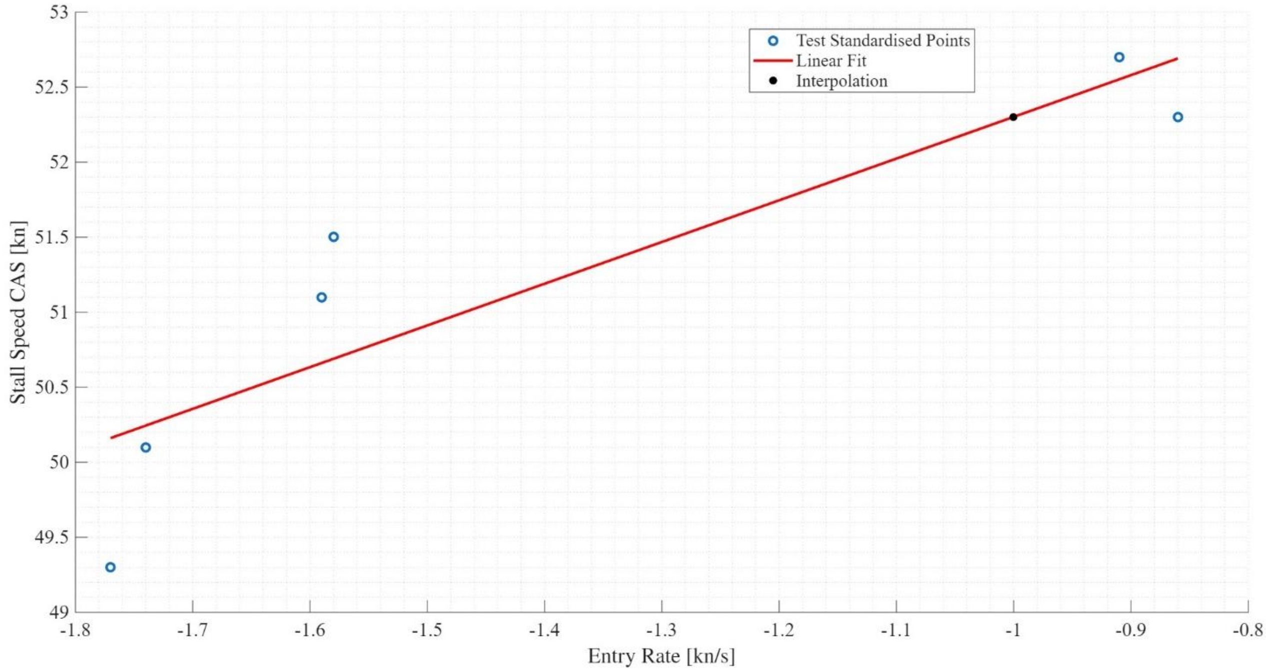

Stall 3
CAS [kt]
GPS Altitude [m]
Stall Point
Entry 1.1 * V$_s$
Stall Warning
Stall
Vertical G [g]
Lateral G [g]
Pitch [deg]
Pitch Rate [deg/s]
Roll [deg]
Roll Rate [deg/s]
Heading [deg]
Yaw Rate [deg/s]
Time [s]


Stall 4
CAS [kt]
GPS Altitude [m]
Stall Point
Entry 1.1 * V$_s$
Stall Warning
Stall
Vertical G [g]
Lateral G [g]
Pitch [deg]
Pitch Rate [deg/s]
Roll [deg]
Roll Rate [deg/s]
Heading [deg]
Yaw Rate [deg/s]
Time [s]


Stall 5
Stall Point
Entry 1.1 * Vs
Stall Warning
Stall
CAS [kt]
GPS Altitude [m]
Vertical G [g]
Lateral G [g]
Pitch [deg]
Pitch Rate [deg/s]
Roll [deg]
Roll Rate [deg/s]
Heading [deg]
Yaw Rate [deg/s]
Time [s]

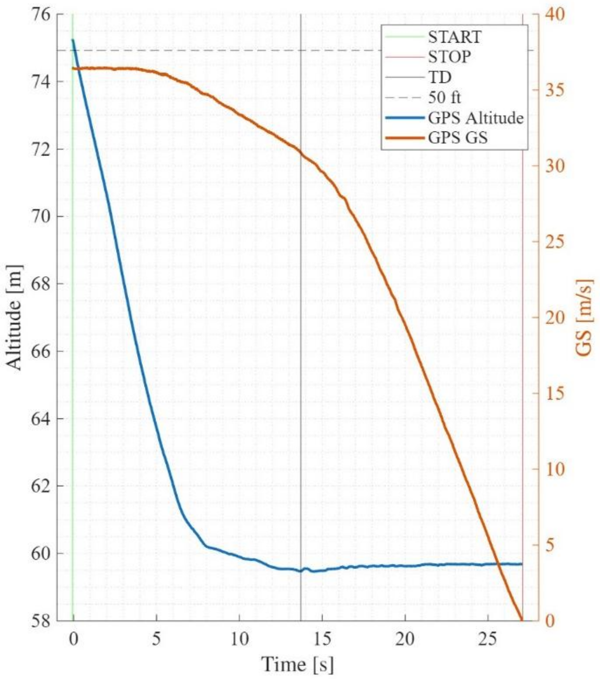

Stall 6
Stall Point
Entry 1.1 * Vs
Stall Warning
Stall
CAS [kt]
GPS Altitude [m]
Vertical G [g]
Lateral G [g]
Pitch [deg]
Pitch Rate [deg/s]
Roll [deg]
Roll Rate [deg/s]
Heading [deg]
Yaw Rate [deg/s]
Time [s]

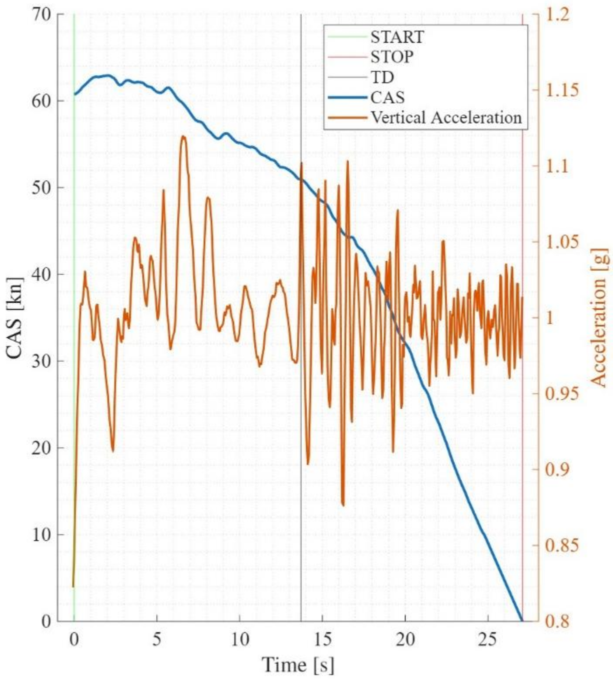
Linear Interpolation of stall results



Landing Performance

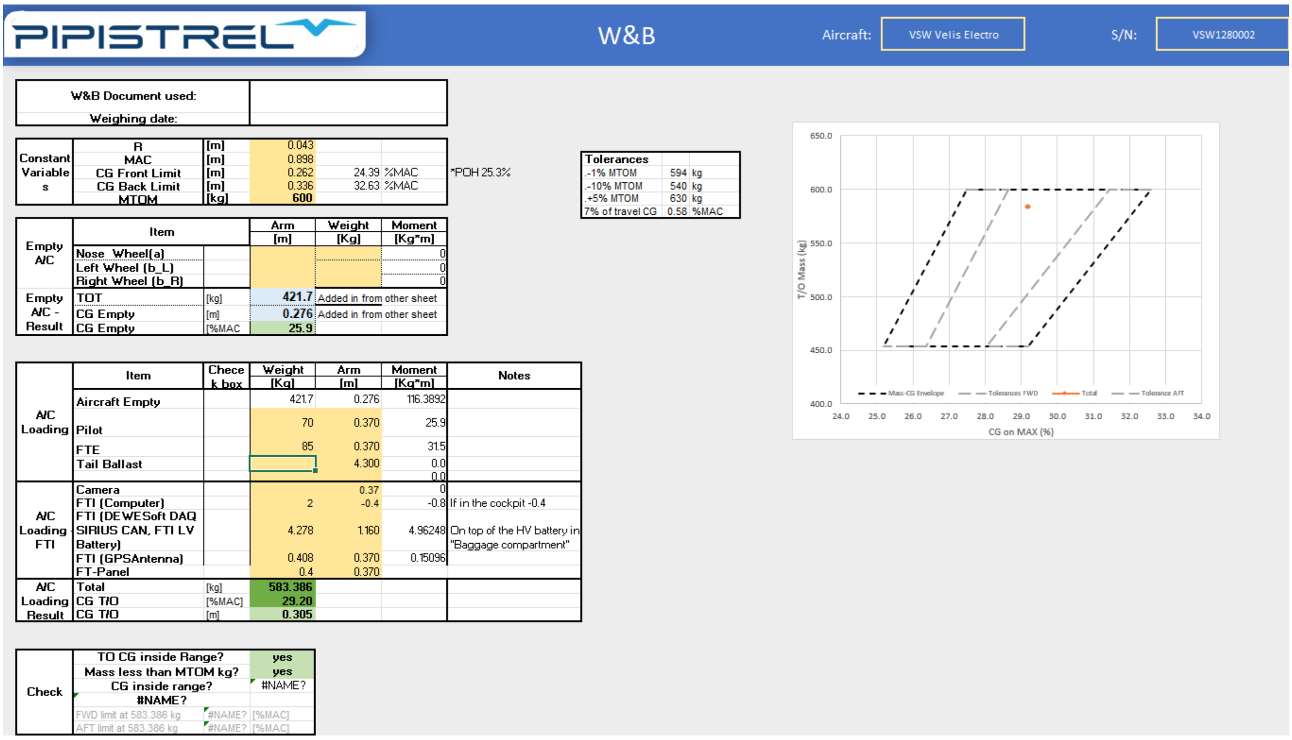
Landing manoeuvre development of speeds and altitude.





Weight & Balance Table

Following a picture of the Weight and Balance charts provided by the company (PVS) and used for the data regression and analysis.



W&B Aircraft: VSW Velis Electro S/N: VSW1280002

W&B Document used:	
Weighing date:	

Constant Variables	R	[m]	0.043			
	MAC	[m]	0.898			
	CG Front Limit	[m]	0.262	24.39	%MAC	*POH 25.3%
	CG Back Limit	[m]	0.336	32.63	%MAC	
	MTOM	[kg]	600			

Tolerances		
.-1% MTOM	594	kg
.-10% MTOM	540	kg
.+5% MTOM	630	kg
7% of travel CG	0.58	%MAC

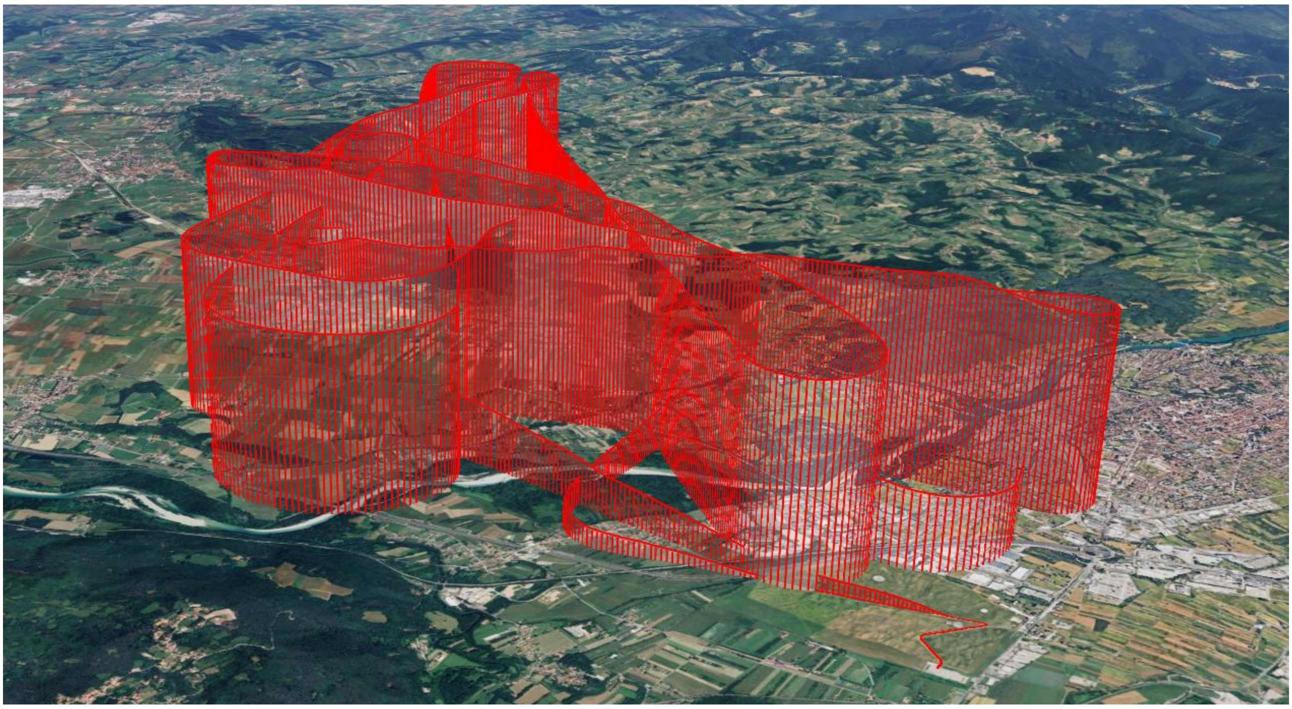
	Item		Arm [m]	Weight [Kg]	Moment [Kg*m]
Empty A/C	Nose Wheel(a)				0
	Left Wheel (b_L)				0
	Right Wheel (b_R)				0
Empty A/C - Result	TOT	[kg]	421.7	Added in from other sheet	
	CG Empty	[m]	0.276	Added in from other sheet	
	CG Empty	[%MAC	25.9		

	Item	Checek box	Weight [Kg]	Arm [m]	Moment [Kg*m]	Notes
A/C Loading	Aircraft Empty		421.7	0.276	116.3892	
	Pilot		70	0.370	25.9	
	FTE		85	0.370	31.5	
	Tail Ballast			4.300	0.0	
					0.0	
A/C Loading FTI	Camera			0.37	0	
	FTI (Computer)		2	-0.4	-0.8	If in the cockpit -0.4
	FTI (DEWESoft DAQ SIRIUS CAN, FTI LV Battery)		4.278	1.160	4.96248	On top of the HV battery in "Baggage compartment"
	FTI (GPSAntenna)		0.408	0.370	0.15096	
	FT-Panel		0.4	0.370		
A/C Loading Result	Total	[kg]	583.386			
	CG T/O	[%MAC]	29.20			
	CG T/O	[m]	0.305			

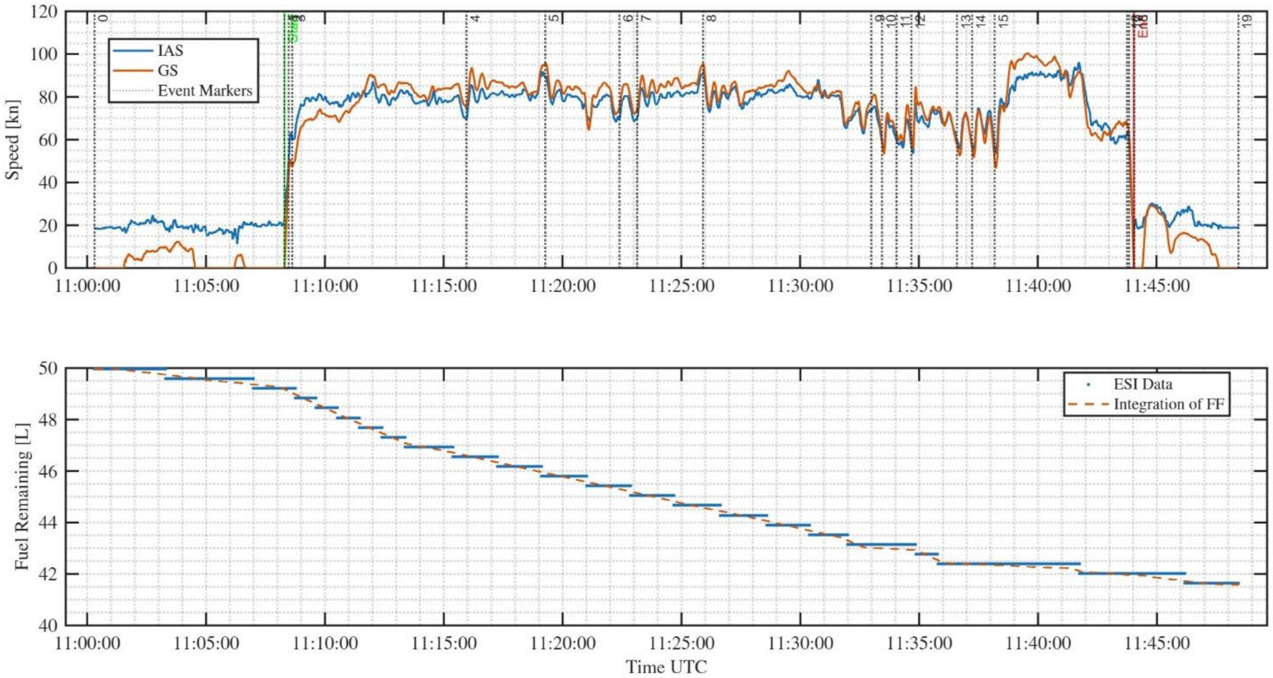
Check	TO CG inside Range?		yes
	Mass less than MTOM kg?		yes
	CG inside range?		#NAME?
	#NAME?		
	FWD limit at 583.386 kg	#NAME?	[%MAC]
	AFT limit at 583.386 kg	#NAME?	[%MAC]

SW121 – Explorer

The figure shows the 3D flight trajectory of the SW121.

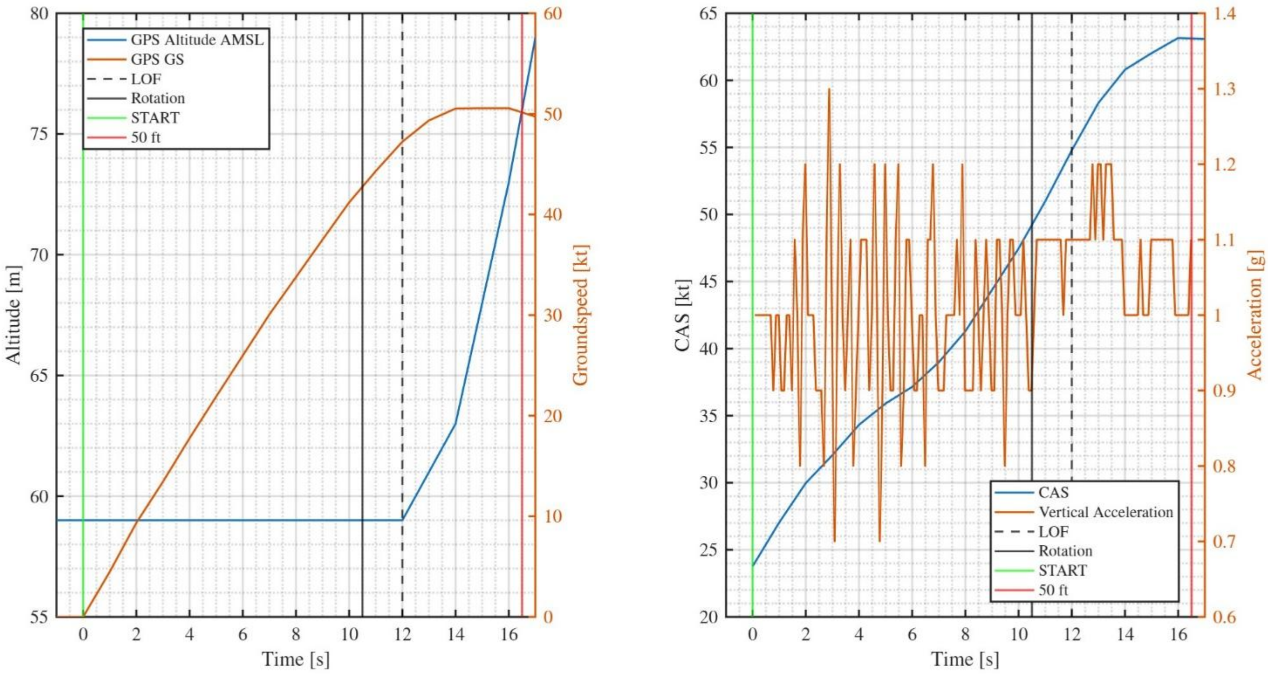


The figure shows the IAS and GS during the SW121 test mission.



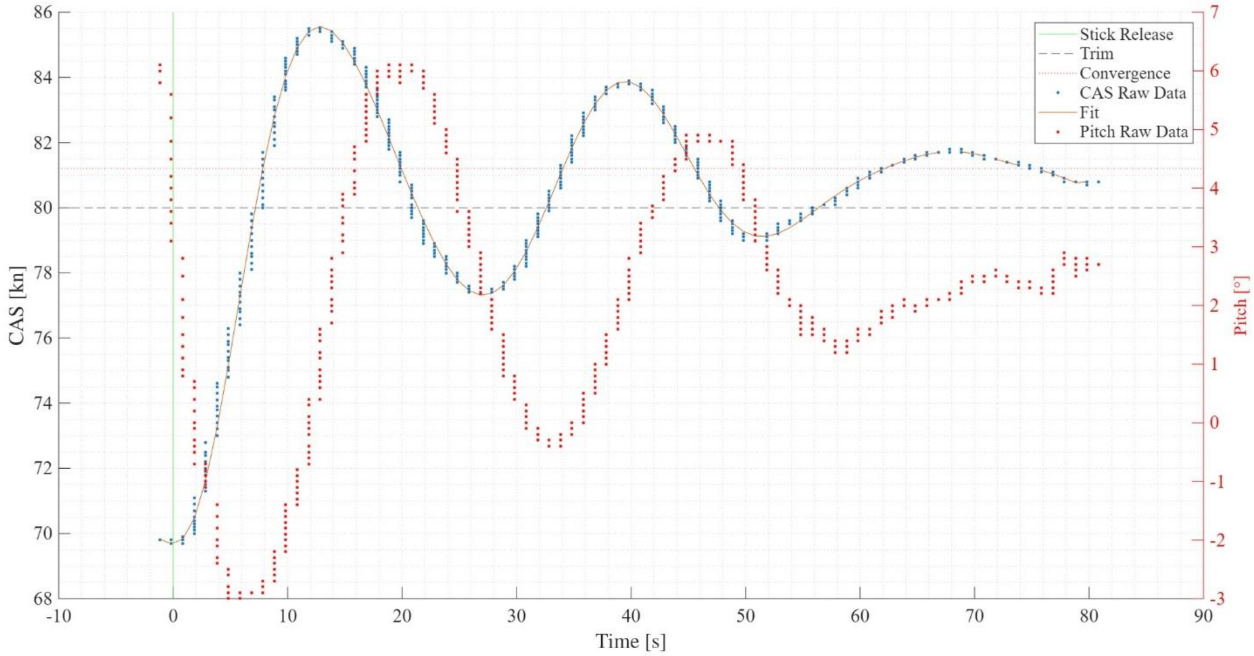
Take-off Performance

Take-off manoeuvre development in time.

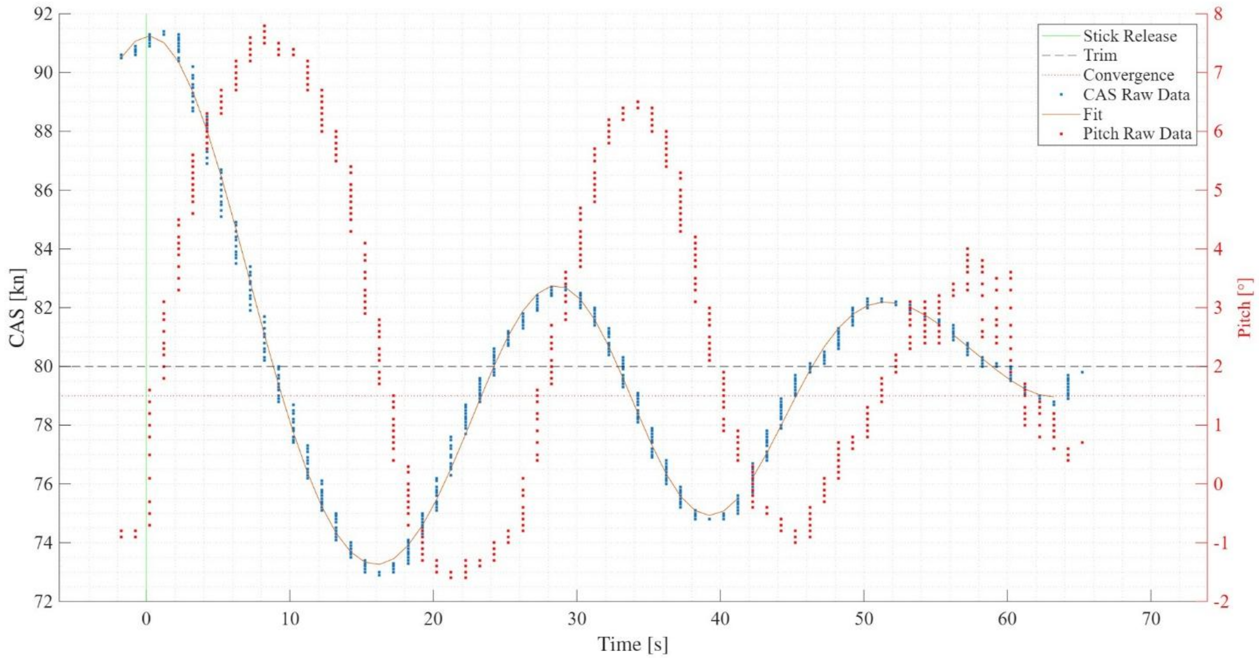


Longitudinal Dynamic Stability – Phugoid

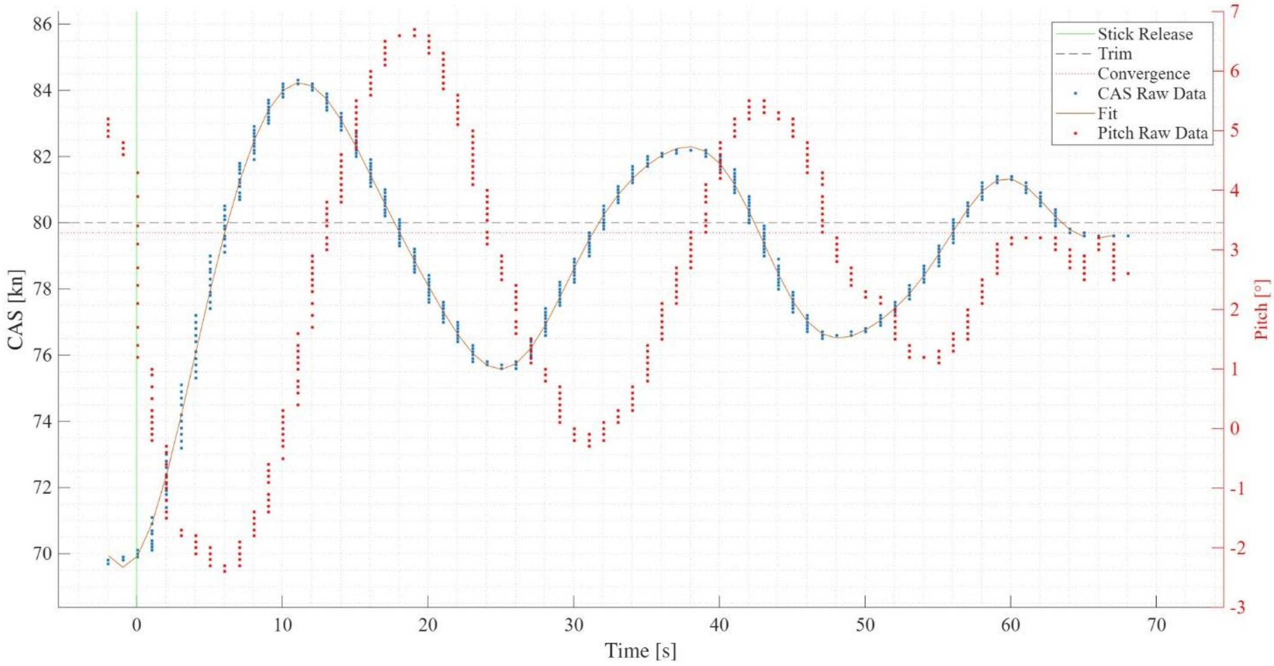
Stick – Free first test point.



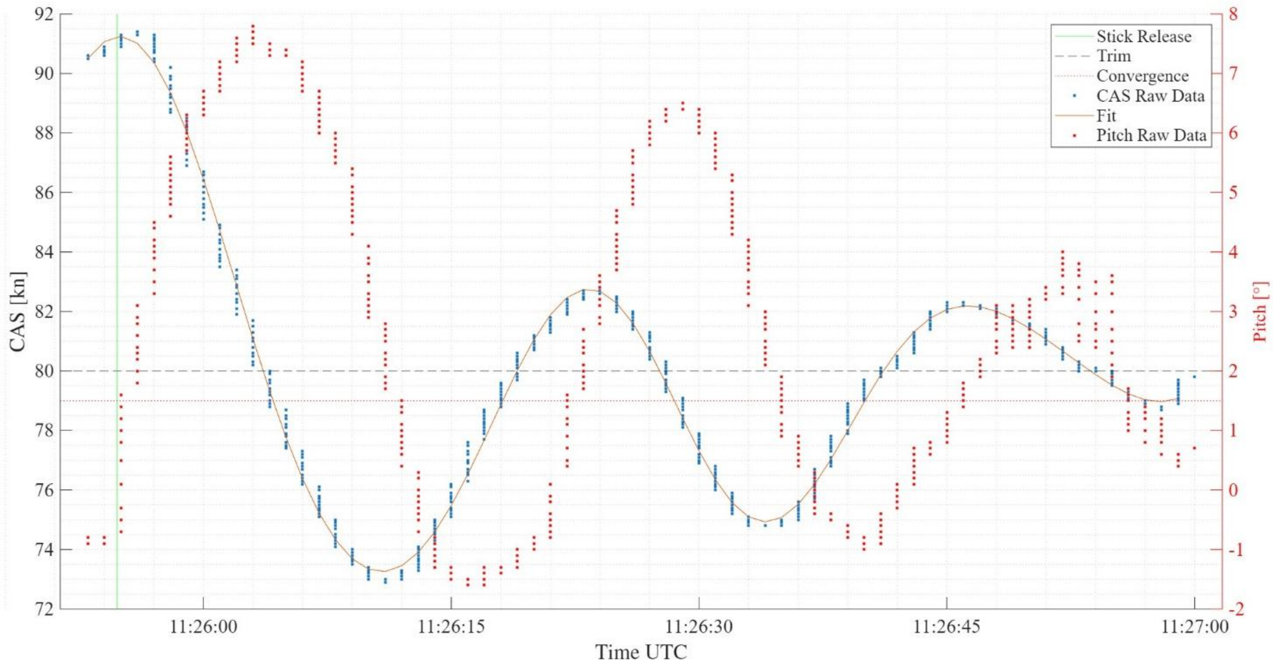
Stick – Free second test point.



Stick – Fixed first test point.

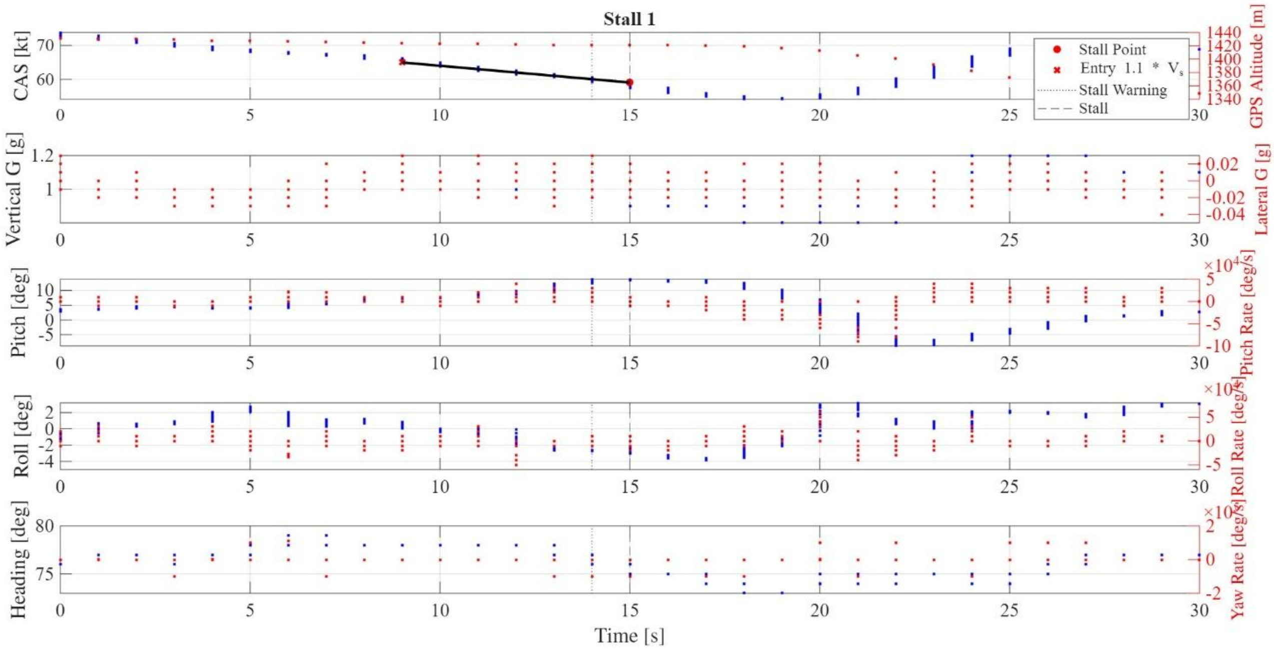


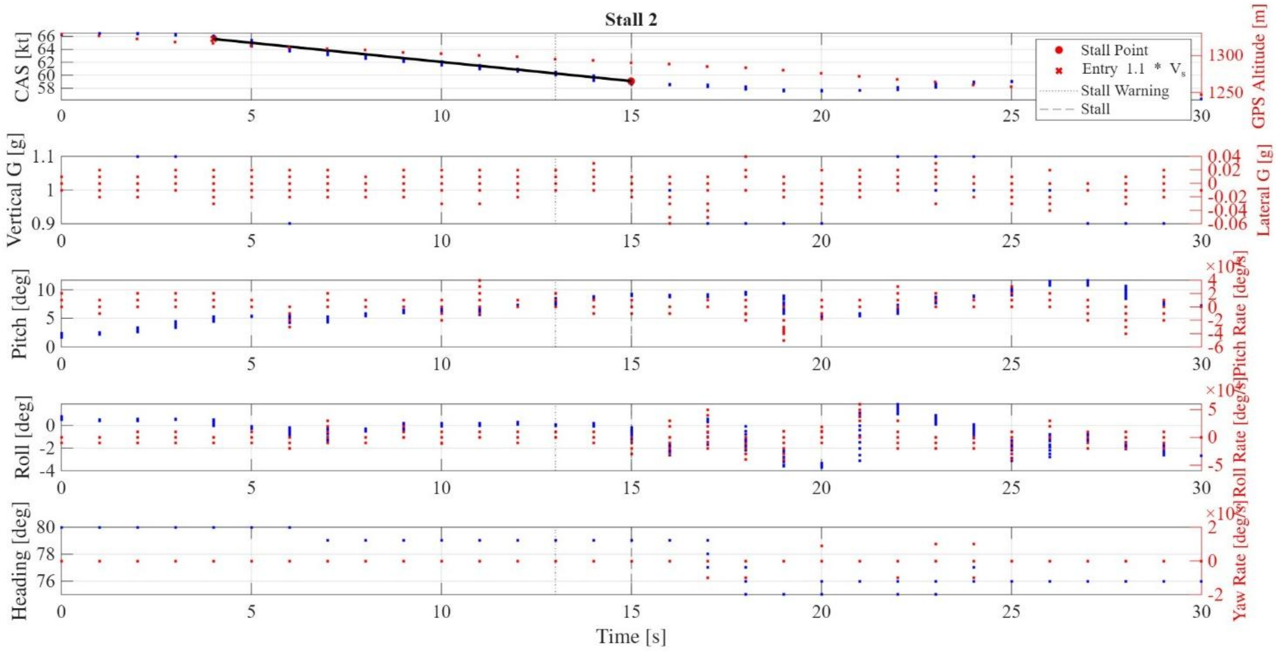
Stick – Fixed second test point.

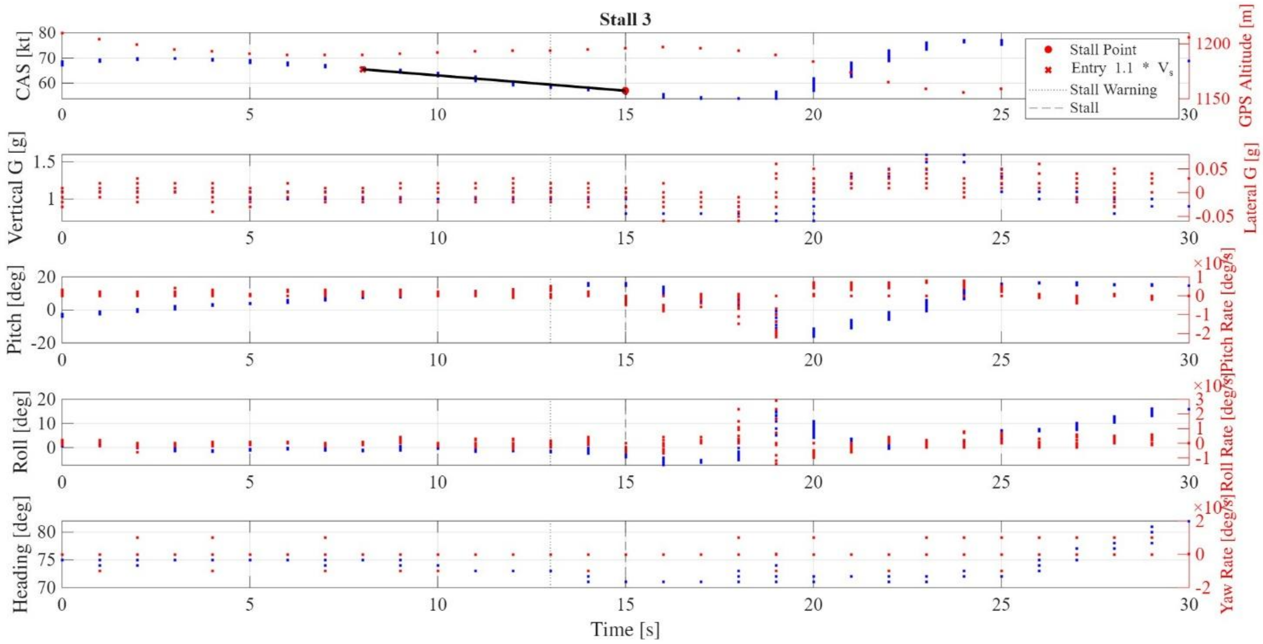


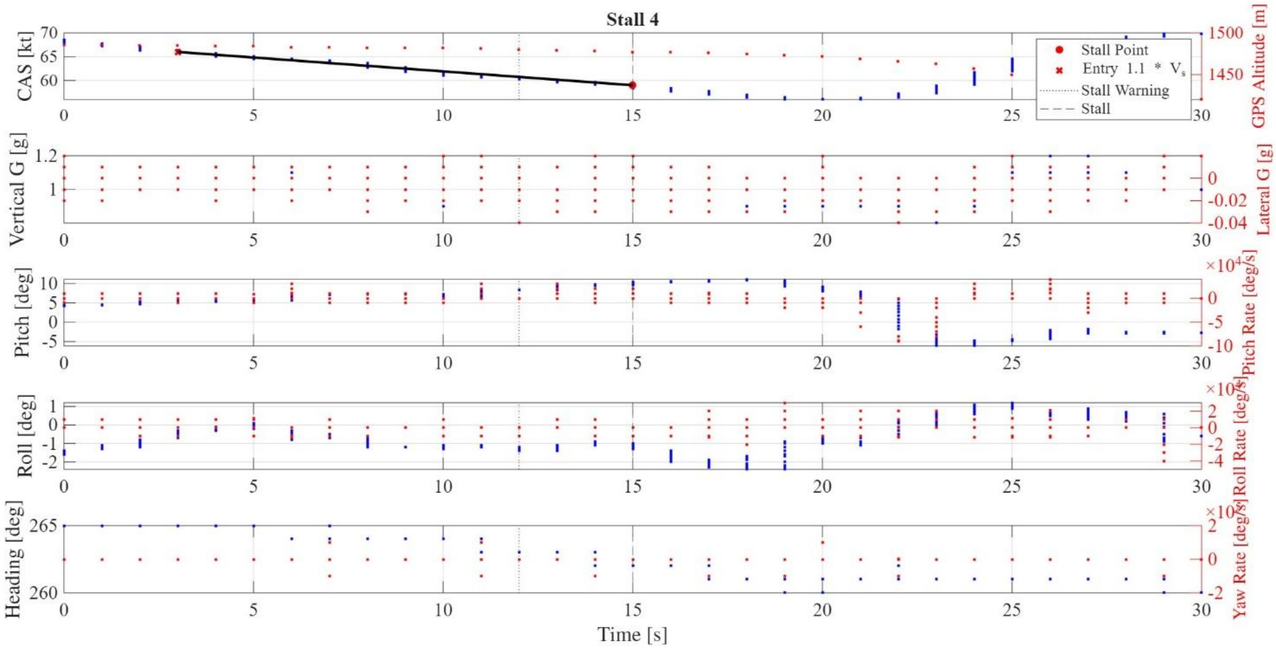
Stall Performance

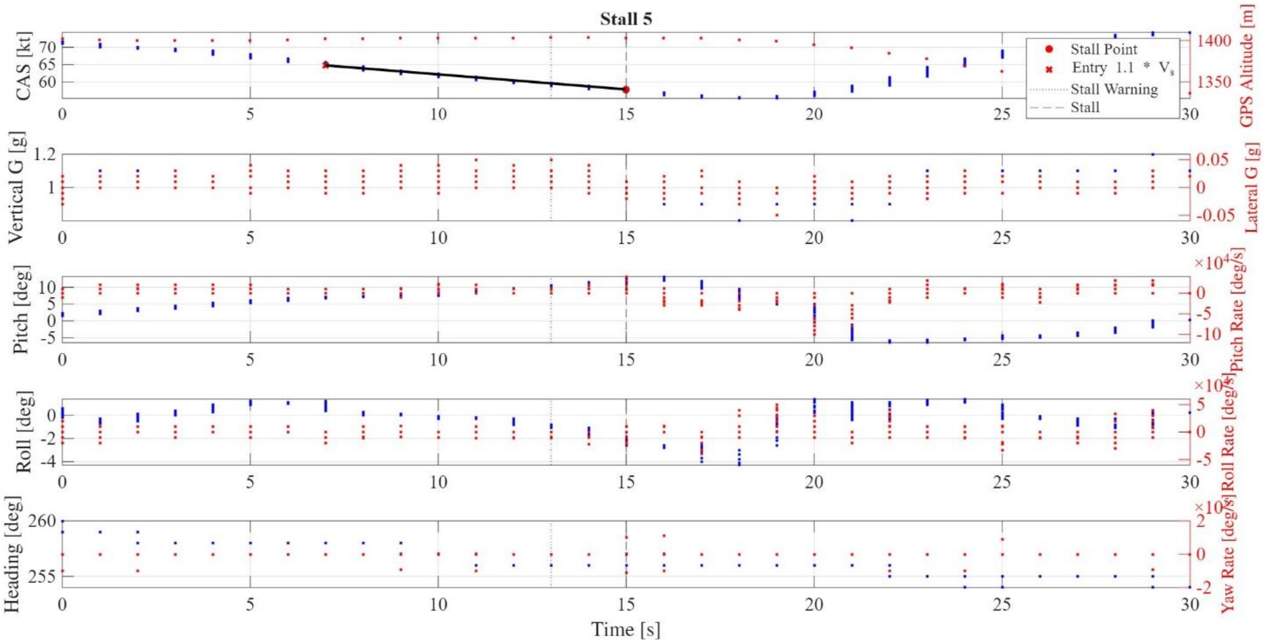
Gathering of all the SW121 stall test points.

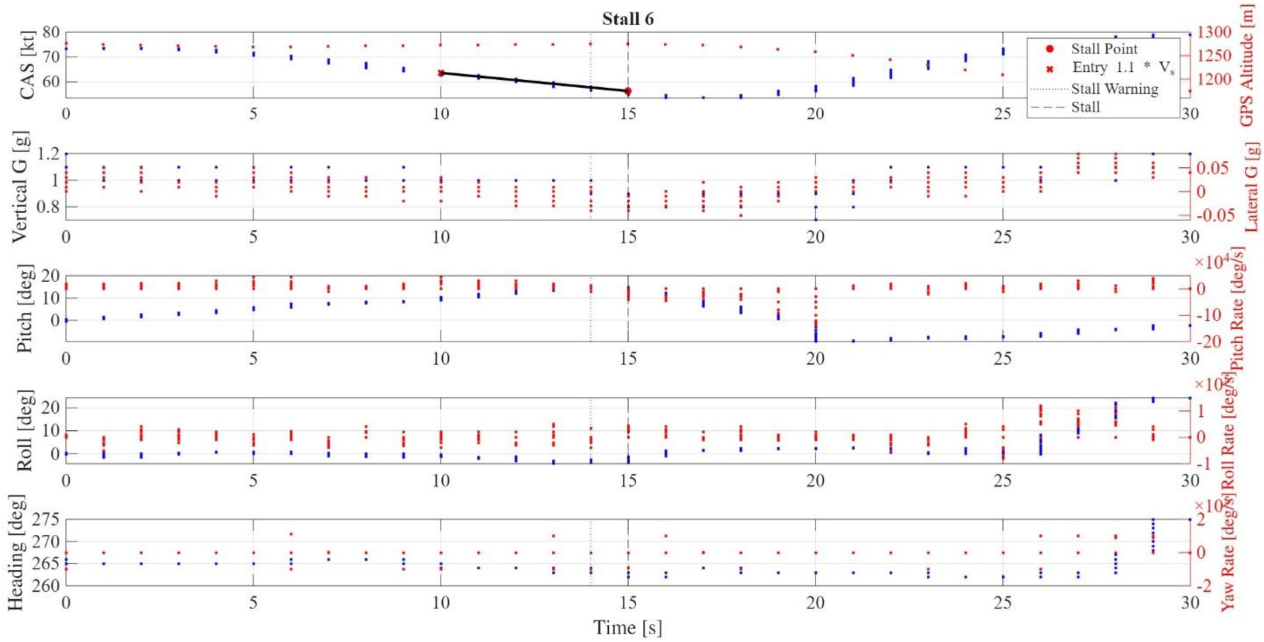




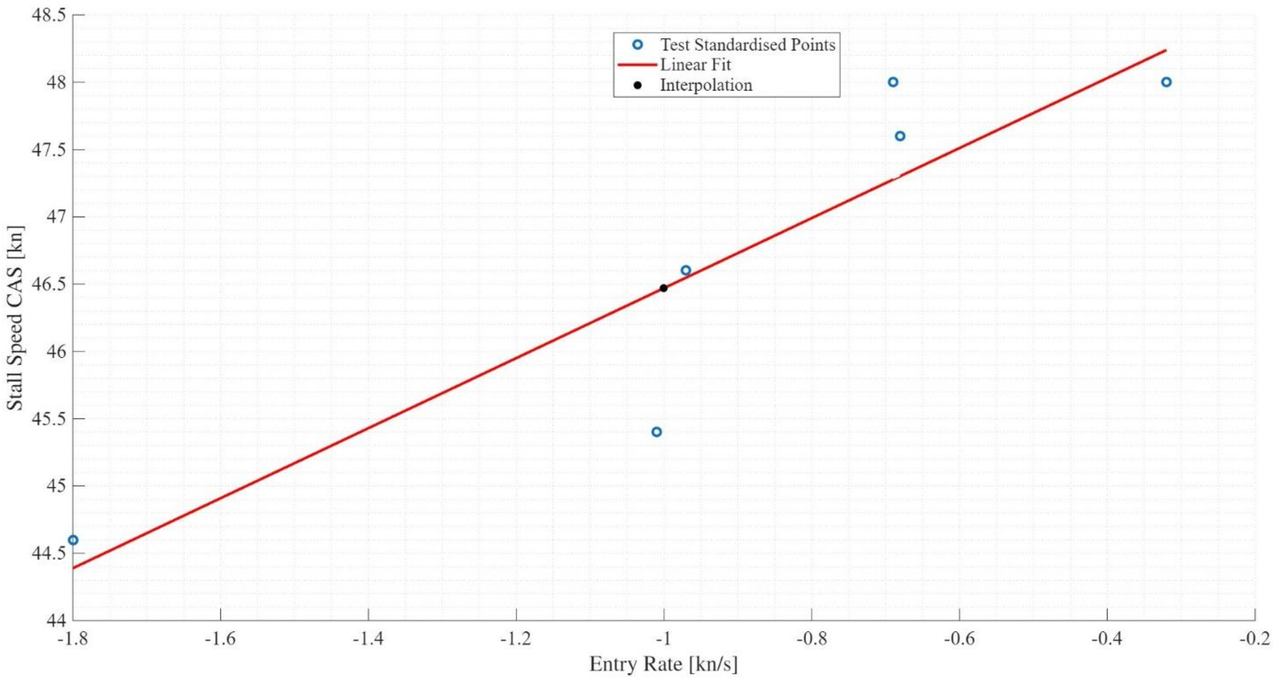

Stall 3
Stall Point
Entry 1.1 * Vs
Stall Warning
Stall
CAS [kt]
GPS Altitude [m]
Vertical G [g]
Lateral G [g]
Pitch [deg]
Pitch Rate [deg/s]
Roll [deg]
Roll Rate [deg/s]
Heading [deg]
Yaw Rate [deg/s]
Time [s]


Stall 4
Stall Point
Entry 1.1 * Vs
Stall Warning
Stall
CAS [kt]
GPS Altitude [m]
Vertical G [g]
Lateral G [g]
Pitch [deg]
Pitch Rate [deg/s]
Roll [deg]
Roll Rate [deg/s]
Heading [deg]
Yaw Rate [deg/s]
Time [s]


Stall 5
CAS [kt]
GPS Altitude [m]
Stall Point
Entry 1.1 * Vs
Stall Warning
Stall
Vertical G [g]
Lateral G [g]
Pitch [deg]
Pitch Rate [deg/s]
Roll [deg]
Roll Rate [deg/s]
Heading [deg]
Yaw Rate [deg/s]
Time [s]

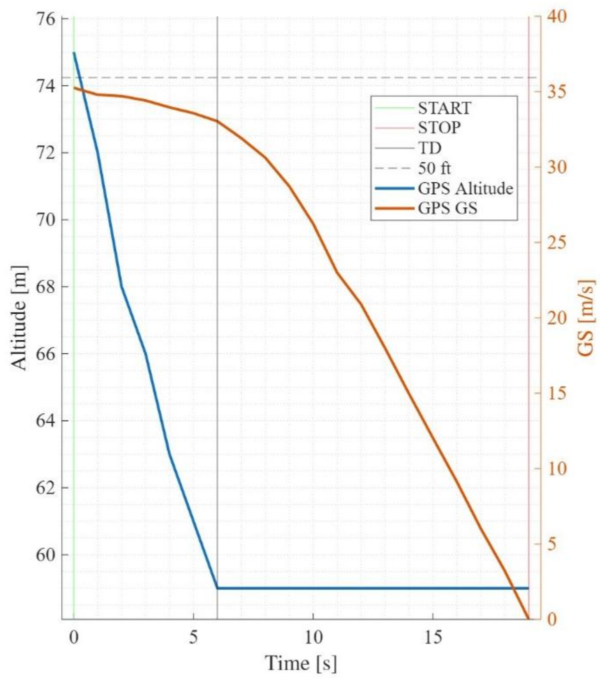

Stall 6
CAS [kt]
GPS Altitude [m]
Stall Point
Entry 1.1 * Vs
Stall Warning
Stall
Vertical G [g]
Lateral G [g]
Pitch [deg]
Pitch Rate [deg/s]
Roll [deg]
Roll Rate [deg/s]
Heading [deg]
Yaw Rate [deg/s]
Time [s]

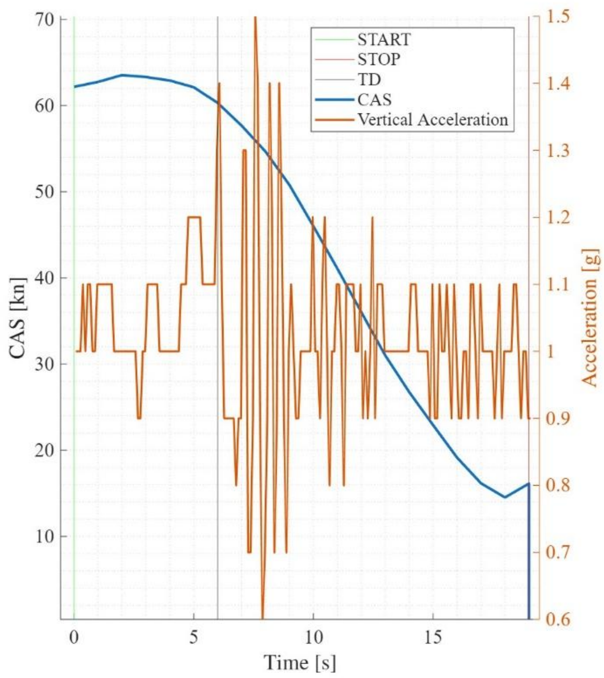
Linear interpolation.



Landing Performance

Landing manoeuvre development of speeds and altitude.





Weight & Balance Table

Following a picture of the Weight and Balance charts provided by the company (PVS) and used for the data regression and analysis.

PIPISTREL VERTICAL SOLUTIONS

Virus SW LSA 121A W&B

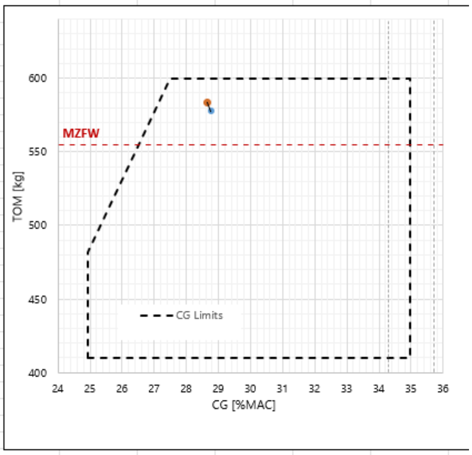
Empty A/C W&B report WBR-121-08-10-0091	S/N	VSW121A0071		
	Weighing date	24/02/2022		
		Weight [Kg]	Arm [m]	Moment [Kg*m]
	LH	158.3	0.491	77.7253
	RH	163.2	0.491	80.1312
	Front	52.3	-1.003	-52.4569
	TOT	**373.8**	-	**105.3996**
	CG Empty [m]	**0.282**		
	CG Empty (%MAC)	**26.64**		

A/C Loading		Weight [Kg]	Arm [m]	Moment [Kg*m]
	Aircraft Empty	373.8	0.282	105.3996
	Parachute and helmet	0	0.370	0
	Pilot	85	0.370	31.5
	FTE	75	0.370	27.75
	Fuel	50	0.215	10.75
	Baggage	0	1.160	0
	Tail Ballast	0	4.300	0
	TOT	**583.80000**	-	**175.3**
	CG T/O [m]	**0.300**		Range [267-375]mm
	CG T/O (%MAC)	**28.66**		Range [25%-37%]

Mission data	Mission Duration [min]	48
	Fuel consumption [L/h]	10.4341
	Fuel burnt [L]	8.34728
	Fuel burnt [kg]	6.0100416
	Fuel at landing	41.65272
	Endurance at landing [min]	240
	Landing Mass	**577.7899584**
	CG LDG [m]	0.30
	CG LDG [%MAC]	**28.76**

Check	Datum	0	
	Front limit	0.267	25%
	Aft limit	0.375	37%
	Fuel sufficient	yes	30
	Weight inside Range?	yes	600
	CG inside Range?	yes	

Fuel quantity [L]	
	50
max 100L	



Teacher's comments